

# Supplement to Fair Outcomes, Unfair Rules: Higher-Order Metamorphic Testing for Procedural Fairness

ANONYMOUS AUTHOR(S)

This supplement gives the protocols behind the paper’s evaluation and the supporting results for each of its four research questions. The protocol sections specify the tests, the training of interaction steering and of the comparison methods, and the statistical analysis. The result sections follow the paper: attribute-flip checks on every model and learning controls for testing (RQ1); the stability of the localization and the controls of repair from failing tests (RQ2); variation across runs, the location of the utility cost, group error rates, joint attributes, and models that pass equality by construction for repair (RQ3); and further edits, populations, updates, and sensitivity analyses for regression and robustness (RQ4). The last results section reports methods with a different prediction interface or a smaller evaluation scope. The replication package supplies per-seed measurements and statistical tests.

## S1 How to use this supplement

This supplement gives the protocols and supporting results for the paper. It uses the paper’s vocabulary and notation, which Section S2.4 restates, and labels the models repaired by interaction steering INTERFAIR.

Sections S2–S4 give the common protocol. Sections S5–S8 follow the paper’s four research questions, and Table S1 shows where each part of the paper is supported. Section S9 reports methods with a different prediction interface or a smaller evaluation scope, including NeuFair. The tables add uncertainty, controls, or results the paper defers; the MLP comparison remains in the main paper, and Table S15 gives the two FT-Transformer settings.

## S2 Data, specifications, and measurements

### S2.1 Tasks and data partitions

Table S2 gives the data partitions. HMDA predicts the recorded approval of a mortgage application. Credit predicts non-default. ACSIncome predicts annual income above \$50,000, and ACSEmployment predicts civilian employment at work. We orient every binary label so that 1 denotes the favorable outcome. The main architectures are a multilayer perceptron (MLP) and FT-Transformer (FT). ERM denotes empirical risk minimization: training with the task loss alone. Each principal comparison uses ten optimizer seeds, 1000–1009, on the same data partition.

The HMDA samples cover Ohio, Maryland, Virginia, and Pennsylvania in 2019. These are independently fitted state models. The transfer study instead takes the Ohio model and applies it without refitting to Michigan 2019 and Ohio 2020. Separating these designs matters: the state comparisons test repeatability of the method, whereas transfer tests a fixed model on another population.

ACSIncome retains respondents older than 16, with annual income above \$100, positive weekly work hours, and person weight at least one. ACSEmployment retains ages 17–89 and person weight at least one, with label ESR=1. Its two settings use sex or an age band (below 40 versus at least 40) as the protected input. Exact age is excluded from the age-band predictor; raw ages 35 and 45 represent the bands when checking edit applicability. Duplicate Credit profiles and census households remain within a single partition. Encoders and edit endpoints are fitted on training records. Calibration records are disjoint from the audit, and the support cutoffs described below are estimated from training records only.

Table S1. Where the supplement supports each part of the paper.

| Paper                           | Supplement                                                                                                                                                                                                                                                                                                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Specification and tests         | Test construction and the linked income edit (Section S2.2); validity checks, support screen, and coverage by group (Section S2.3, Table S3)                                                                                                                                                                             |
| Interaction steering            | Objective and the reason for the outcome terms (Section S3); configuration (Table S5); checkpoint selection (Section S3.2)                                                                                                                                                                                               |
| Experimental setup              | Partitions (Table S2); metrics (Table S4); comparison protocols (Table S6); test families and ranking (Section S4)                                                                                                                                                                                                       |
| RQ1 (Testing)                   | Fairness checks on the running example (Table S9); attribute-flip checks on every model (Table S10); reporting codes (Section S5.3); learning controls (Table S11)                                                                                                                                                       |
| RQ2 (Localization)              | Stability of the localization (Table S12); discovery and the two arms the paper omits (Table S13); replay (Table S14)                                                                                                                                                                                                    |
| RQ3 (Repair)                    | FT-Transformer comparison (Table S15); variation and paired intervals (Tables S16 and S17); score contraction (Table S18); situation testing and its controls (Tables S19–S22); error rates (Table S23); joint attributes (Section S7.6); ablation (Section S7.7); models that pass equality by construction (Table S24) |
| RQ4 (Regression and robustness) | Additional edits (Table S25); FT-Transformer transfer (Tables S26 and S27); maintenance (Table S28); training edits, training pools, and audit size (Section S8.5)                                                                                                                                                       |
| Comparison methods              | MirrorFair, equalized-odds post-processing, DICE, other model families, and NeuFair (Section S9)                                                                                                                                                                                                                         |
| Discussion and threats          | Removal and proxies (Section S7.8); audit size (Section S8.5); other model families (Table S34); test families (Tables S7 and S8)                                                                                                                                                                                        |

Table S2. Partition sizes. Each HMDA state has its own sample and fitted preprocessing. Census tasks use Maryland’s 2018 one-year records.

| Task             | Training | Validation | Calibration | Audit  |
|------------------|----------|------------|-------------|--------|
| HMDA, each state | 39,000   | 7,200      | 1,800       | 12,000 |
| Credit           | 19,495   | 3,601      | 900         | 6,004  |
| ACSIIncome       | 21,482   | 3,963      | 996         | 6,601  |
| ACSEmployment    | 31,138   | 5,788      | 1,455       | 9,576  |

## S2.2 Constructing the tests

An ordinary feature is a semantic field, which may occupy several encoded columns. Changing a category replaces its complete encoding. Missing-value indicators and linked fields are updated together. This avoids tests that accidentally create two active categories, retain an incompatible missingness flag, or alter only part of a dependent quantity.

The default numerical contrast uses the observed training values nearest the 20th and 80th percentiles. Categorical contrasts use the two most frequent admissible training levels. Equality is required for all ordinary fields: 13 in HMDA, 21 in Credit, and five in each census setting. Directions are supplied only where the edit has a stated interpretation: increasing income and decreasing debt-to-income or loan-to-value ratios for HMDA; increasing payment amounts for Credit; increasing education or work hours for ACSIncome; and increasing education for ACSEmployment. An unordered category has no assigned direction.

*A linked income edit.* For mortgage applicants with positive income and an exact reported debt-to-income ratio (DTI) from 36 to 49, we raise annual income while retaining the debt implied by

Table S3. Coverage of the ordinary edit tests. Counts are feature–profile tests; every accepted test includes both protected settings. Natural-record utility and group metrics use the entire audit partition.

| Task / recorded group | Candidates | Schema-valid | Supported | Retained (%) |
|-----------------------|------------|--------------|-----------|--------------|
| HMDA Ohio / White     | 24,830     | 24,326       | 19,040    | 76.68        |
| HMDA Ohio / Black     | 1,794      | 1,758        | 1,319     | 73.52        |
| HMDA Ohio / all       | 26,624     | 26,084       | 20,359    | 76.47        |
| Credit / male         | 16,947     | 16,947       | 15,743    | 92.90        |
| Credit / female       | 26,061     | 26,061       | 24,689    | 94.74        |
| Credit / all          | 43,008     | 43,008       | 40,432    | 94.01        |
| ACSIncome / male      | 5,280      | 5,234        | 4,753     | 90.02        |
| ACSIncome / female    | 4,960      | 4,940        | 4,420     | 89.11        |
| ACSIncome / all       | 10,240     | 10,174       | 9,173     | 89.58        |

those values. If income is  $I$  and the increment is  $h$ , the updated DTI is  $rI/(I + h)$ . The result is mapped back to an HMDA reporting code before prediction. For example, the running example’s DTI changes from 45 to a calculated 31.8, represented by code 33 for the reported 30–36% interval. This mapping is part of the edit, not a model-dependent adjustment. Interaction steering sees  $h = \$10,000$  during training; increments of \$20,000, \$30,000, and \$50,000 test generalization.

The two protected settings crossed with the two edit endpoints give four inputs. Each is run through the complete fitted pipeline, including a baseline’s preprocessing. All methods evaluated on a given specification share the same accepted tests. A failing test stores the profile, the edit, the four inputs and outputs, and the check that failed, so a later model can run exactly the same test.

S2.3 Checking applicability and empirical support

We first check field bounds, category membership, missing values, and declared consistency rules. We then check that all four inputs lie sufficiently close to training records of their respective protected group. The purpose is to avoid attributing a rule violation solely to a query far outside the observed data.

Distance is Euclidean in the fitted input encoding, excluding the protected coordinate. Numerical fields are median-imputed, clipped at the training .005 and .995 quantiles, centered by the median, and divided by the interquartile range. A winsorized standard deviation is the fallback, with a scale floor of 1. Missing indicators and one-hot categories, including an unknown-category indicator, remain in the distance representation.

Within a protected group of size  $n_g$ , a fixed shuffle selects  $\min(1000, \max(64, \lfloor n_g/5 \rfloor))$  training records for distance calibration and up to 6,000 different training records as neighbors. The 95th percentile of calibration records’ fifth-neighbor distances defines that group’s cutoff. A test is retained only if each of its four inputs is within the cutoff of the group its protected value encodes. Thus, a Black-coded input is checked against Black-coded training profiles even when the starting record was White-coded. This is a proximity diagnostic, not a causal model of how a person’s attributes would change.

Each audit starts from 2,048 held-out profiles. Table S3 shows why the number of tests differs from that starting budget: one profile contributes a test for each applicable feature. Rejection also varies by recorded group, so coverage accompanies the failure rates. Section S8.4 compares different training endpoints and the effect of omitting the proximity screen during training.

Table S4. Measurements and units. Lower is better for gaps and failure rates; higher is better for the three utility measures.

| Measure      | Meaning                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| AUROC        | Ranking of favorable versus unfavorable recorded labels, summarized by area under the receiver operating characteristic curve.  |
| Accuracy, F1 | Fraction of correct decisions and the harmonic mean of precision and recall for the favorable label, at threshold .5.           |
| AOD          | Half the sum of the absolute true-positive-rate and false-positive-rate gaps. Opposite signed gaps do not cancel.               |
| DP, EO       | Absolute positive-decision-rate gap; maximum of the absolute true- and false-positive-rate gaps.                                |
| $\bar{R}$    | Mean absolute logit residual, averaged equally over the ordinary edits.                                                         |
| $V$ (%)      | Percentage of ordinary tests with $ R  > .05$ .                                                                                 |
| $D$ (%)      | Percentage with unequal signed decision changes: $d(1, T(x)) - d(1, x) \neq d(0, T(x)) - d(0, x)$ .                             |
| $W$ (%)      | Percentage of signed tests for which at least one protected setting moves in the wrong probability direction by more than .005. |

## S2.4 Metric definitions

Let  $a$  denote the protected setting and  $x$  the ordinary inputs. Write  $z(a, x)$  for the log-odds score,  $p(a, x) = \sigma(z(a, x))$  for its probability under the logistic link, and  $d(a, x) = 1[p(a, x) \geq .5]$  for its decision. An edit  $T$  induces response  $e_a = z(a, T(x)) - z(a, x)$ . The interaction residual is  $R = e_1 - e_0$ . It measures whether the effect of the edit changes with the protected input. Every residual is deterministic for a fixed model and a fixed test.

Table S4 defines the measures used throughout. Behavioral rates are computed within each edit and then averaged equally across edits. This prevents a feature with more supported contexts from dominating the result. The signed-edit rate  $W$  uses only edits for which a direction was supplied. Utility and group measures use observed, unedited audit records.

The equality tolerance .05 is in log-odds; the .005 reporting margin is in probability. Direction training targets a nonnegative signed logit response, without this reporting margin. Section S7.3 also reports wrong-direction signs with zero margin. Equality of logit responses does not imply equal probability increments or decision changes: a constant protected-group offset can put two equal responses on different sides of the decision threshold.

## S3 Training interaction steering and comparison methods

### S3.1 The training objective

Interaction steering starts from random weights and adds the specification’s penalties to binary cross-entropy on observed labels. Edited inputs supply comparisons between model outputs, not invented task labels. There are four terms: the rule terms, equality  $\mathcal{L}_{\text{eq}}$  and direction  $\mathcal{L}_{\text{dir}}$ , and the outcome terms, population  $\mathcal{L}_{\text{pop}}$  and decision  $\mathcal{L}_{\text{dec}}$ . The next paragraph gives the reason for the outcome terms; the rest of this section specifies each term.

*Why outcome terms, and why edited inputs carry no labels.* Steering constrains *how* the model may use a factor while it still learns the task; it does not prescribe outputs. Without outcome terms, the optimizer can take degenerate paths to a small finite difference, such as flattening the protected pathway or driving all four executions to the same value, and a repaired rule need not narrow outcome gaps. Besides the task loss, we use two outcome terms: the population term  $\mathcal{L}_{\text{pop}}$  penalizes differences between the groups’ mean scores and mean edit responses, and the decision term  $\mathcal{L}_{\text{dec}}$

penalizes differences in the two values' smoothed decision responses and in group decision rates. The task loss is computed only on natural, labeled examples. Edited inputs appear only inside the penalties, and none carries a label. Unlike counterfactual data augmentation, which trains on edited inputs [6], our method uses edited inputs only as points at which effects are read, so the same frozen tests can audit any model after the fact, including baselines that never saw it.

For a vector of residuals  $v$ , the equality penalty uses

$$H(v) = \frac{1}{n} \sum_i v_i^2 + \frac{.25}{k} \sum_{i \in K(v)} v_i^2, \quad k = \lceil .1n \rceil, \quad (S1)$$

where  $K(v)$  contains the largest  $k$  squared residuals. At each update a relation, one edit with its check, is sampled uniformly, and the loss is estimated on its *contexts*: supported training profiles to which the edit applies. The first term reduces average failure; the second gives extra weight to severe failures. Equality is averaged separately over ordinary and signed relations and the two averages are added. An edit in both collections consequently contributes to both averages. For a signed edit with required sign  $s_T$ , the direction penalty applies  $H$  to  $\max(0, -s_T e_a)$ , over contexts and protected settings.

*Population scores and effects.* For values  $v_i$  on observed records, let  $Q(v)$  average squared differences between the protected groups' means in three scopes: all records, recorded label 0, and recorded label 1. A scope contributes only when it has at least eight records in each group; eligible scopes have equal weight. If none is eligible, the penalty is zero. The population objective applies  $Q$  to natural-record logits and to each edit's logit response under the record's observed protected value. The effect penalties average features equally.

A boundary-weighted version gives more attention to records near the decision threshold. It replaces cell means by weighted means, using  $w_i = \exp(-|p_i - .5|/b)$  with the weights detached from differentiation and normalized by their batch mean. The bandwidth  $b$  is .10 for HMDA and .18 for the other tasks. Denominators are bounded below by  $10^{-6}$ ; the eight-record rule still applies. Unweighted score and effect penalties have coefficients .2 and .4. Their weighted counterparts have coefficients .2 and .3 for HMDA, and .15 and .25 otherwise.

*Decision responses and rates.* Training approximates a threshold decision by  $q_\tau(z) = \sigma(z/\tau)$ . The response term penalizes the squared difference in  $q_\tau(z(a, T(x))) - q_\tau(z(a, x))$  between protected settings, averaged over profiles. The rate term penalizes squared differences between groups' mean  $q_\tau$  values, within each label and unconditionally, with weight .5 on the unconditional gap and the same eight-record eligibility rule. Averaging over several temperatures supplies gradients both close to and farther from the threshold. The response coefficient is 4 with temperatures (.15, .35, .7); the rate coefficient is 16 with temperatures (.25, .5, 1). Evaluation always uses the hard threshold .5 for  $D$ , AOD, DP, and EO.

Each task update also samples 512 natural records for population scores, 256 supported contexts for one ordinary relation, and 256 for one signed relation. Context sampling balances the nonempty protected-by-label cells, drawing with replacement only when a cell is too small. Remaining batch positions are sampled from the eligible pool. Task passes retain the architecture's training behavior. Regularizer passes disable dropout while retaining gradients. FT-Transformer uses numerical inputs and missingness indicators as numerical tokens and converts categorical blocks to category identifiers; the protected input is a binary categorical token.

### S3.2 Validation and checkpoint selection

All selection uses validation records and validation test banks. A candidate is admissible when AUROC, accuracy, and F1 are each within .02 of paired ERM, and when AOD, DP, EO, both

Table S5. Training configuration. Fairness terms warm up linearly over the first five epochs.

| Setting                     | MLP                                                              | FT-Transformer                                                                       |
|-----------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Architecture                | Three hidden layers of 128 ReLU units; scalar output; no dropout | Three blocks; token width 192; eight heads; ReGLU; feed-forward width multiplier 4/3 |
| Dropout                     | None                                                             | Attention .2; feed-forward .1; residual 0                                            |
| Optimizer                   | AdamW, rate .001, decay .0001                                    | AdamW, rate .0001, decay .00001; no decay on embeddings, normalization, or biases    |
| Training budget             | 40 epochs; validation every five epochs                          | At most 100 epochs; validation every epoch; patience 16                              |
| Task batch / clipping       | 512 records / norm 5                                             | 512 records / norm 5                                                                 |
| Equality coefficient        | 1 on HMDA; .35 otherwise                                         | 1 on HMDA; .35 otherwise                                                             |
| Direction coefficient       | 1                                                                | 1                                                                                    |
| Overall fairness multiplier | 1                                                                | .3 on HMDA; 1 on ACSIncome                                                           |

component error-rate gaps,  $D$ , and  $\bar{R}$  are no worse than ERM. Joint race–sex fitting additionally checks sex-marginal and worst-group-pair outcomes. All runs remain in the evaluation.

Candidates are ordered first by admissibility, then by total normalized constraint excess, then by  $V + W + D + .1(\text{AOD} + \text{DP} + \text{EO})$ , and finally by validation cross-entropy. Rates in this expression are fractions. Outcome and decision excesses are divided by their ERM value plus .01; utility excess beyond the .02 allowance is divided by .02; residual excess is divided by  $\max(\bar{R}_{\text{ERM}}, .01)$ . ERM instead selects maximum validation AUROC, breaking ties by cross-entropy. Section S6 isolates the effect of using INTERFAIR’s checkpoint selector without its training penalty.

### S3.3 Comparison methods and their references

We preserve the published methods’ objectives and method-specific stages. Generic classifier fitting uses the common MLP or FT architecture; preprocessing, rationale alignment, mirror-model fitting, and neuron search remain part of the method being evaluated. Table S6 summarizes the operational differences. The repository records full configurations and source versions for each fitted pipeline.

The main tables answer two different comparison questions. The repair tables compare delivered models with a common ERM. RQ1 instead asks what each published pipeline changed relative to its own unmitigated reference. For example, the MLP source used by the LTDD, CoT-Phi, DRAlign, and NeuFair comparisons on the three principal tasks validates every epoch and omits gradient clipping; the primary INTERFAIR/ERM protocol validates every five epochs and clips at 5. The RQ1 paired tests use the corresponding source reference. HIFI’s marked comparisons on those three tasks use the shared reference, as identified in the paper’s table.

Fair-SMOTE and reweighing also have a matched preprocessing comparison with their own ERM, identical initialization, and validation-AUROC selection. Classifiers train for 40 epochs; Fair-SMOTE’s enlarged dataset entails more updates per epoch. The Fair-SMOTE Maryland, Virginia, and Pennsylvania comparisons use AdamW at .0007, decay .0001, batch 512, and norm clipping 5 for both method and reference. Their numerical generation uses training-median imputation and training min/max scaling; categorical fields use the generator’s categorical branch. The situation filter uses logistic regression with  $C = 1$ , L2 regularization, the liblinear solver, and 100 maximum iterations. Its training-data filter is distinct from the label-cohort analysis in Section S7.4.

Table S6. Comparison protocols. Every test queries the complete fitted pipeline at all four inputs.

| Method          | Fitting and evaluation                                                                                                                                                                                                                   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTDD [8]        | Regress encoded coordinates on the protected input and residualize those with $p < .05$ . Mask the protected coordinate in the downstream classifier. Audit residualization and classifier together.                                     |
| CoT-Phi [14]    | Apply the released closed-form correlation parameter and protected-column mutation to training data, keeping labels. Query the fitted classifier on untransformed test inputs.                                                           |
| DRAlign [7]     | Continue its ERM source for 20 epochs with Adam at .001, batches of 1,000 per protected group, endpoint weight .3 and rationale weight .03. Select minimum validation soft DP over epochs 4–20.                                          |
| Reweighting [5] | Weight an observed record by $P(A)P(Y)/P(A, Y)$ ; fit with weighted task loss and validation-AUROC selection.                                                                                                                            |
| Fair-SMOTE [1]  | Generate within protected-by-label cells, using three neighbors and $F = CR = .8$ , then apply the released logistic-regression situation filter. Preserve the generator’s numeric and categorical operations through the input adapter. |
| HIFI [16]       | Retain the released sensitive-only coalition objective, with weight .1. Report shared fitting and the native NAdam/20-epoch schedule separately where both are evaluated.                                                                |
| NeuFair [2]     | Search neuron masks for one hour per seed, with 25,000 uphill-initialization iterations, mask bounds 2–75, and an F1 floor .98 times the source validation F1. Audit the selected masked network.                                        |

All six published methods are fitted on both FT settings and both ACSEmployment tasks. LTDD residualizes encoded coordinates before the FT tokenizer; its categorical argmax assignments remain unchanged on the training records in both tasks, so tokenization discards the continuous corrections to one-hot columns while retaining numerical-coordinate residualization. The protected coordinate is then masked in the classifier, and every audit queries the full transform–classifier pipeline. Fair-SMOTE uses the same generator and situation filter before FT fitting with the common FT optimizer and stopping rule. DRAlign uses its original rationale objective over the FT weights, its original batch size and validation criterion, and the paired main-table FT ERM as its source. NeuFair’s one-hour search remains confined to HMDA Ohio MLP.

#### S4 Statistical analysis

The unit of paired inference is the optimizer seed on a fixed data partition. A paired difference compares methods with the same seed within the relevant protocol; it does not turn two differently trained references into one control. We use two-sided paired  $t$ -tests and apply Holm correction within the comparison families. Each effect is also available as a difference of means and a ratio of means; the latter is the method mean divided by the reference mean, not the average of per-seed ratios. An algebraically zero reference has no defined ratio.

Tables S7 and S8 identify the families supporting the paper’s comparisons. Supporting experiments retain the full families defined for their scientific questions, even when this supplement displays only a subset of their metrics. Exact contrast lists, effect sizes, intervals, and adjusted  $p$ -values are in the machine-readable exports. We do not recompute a smaller family around the rows selected for presentation.

Scott–Knott ESD 2.0.3 [12] groups compatible method outcomes within each setting and metric. Superscripts denote the groups (lower is better), and bold marks the best group among the displayed methods. Ranks retain the complete comparison family, so numbers can be skipped; when grouping is undefined, as for comparisons that are all zero, bold marks the best observed mean without a rank.

Table S7. Principal inferential families. Counts include all tests in a family, not only contrasts mentioned in the text.

| Question / comparison                                                                                | Tests     |
|------------------------------------------------------------------------------------------------------|-----------|
| INTERFAIR versus common ERM, seven task–architecture settings: response / utility and group outcomes | 70 / 42   |
| INTERFAIR versus published methods, those settings: response / utility and group outcomes            | 356 / 225 |
| Published methods versus common ERM, those settings: response / utility and group outcomes           | 296 / 189 |
| Published methods versus their references in the principal RQ1 study                                 | 63        |
| Fair-SMOTE and reweighing, matched preprocessing study: response / utility and group outcomes        | 60 / 36   |
| Maryland, Virginia, Pennsylvania versus common ERM: response / utility and group outcomes            | 126 / 108 |
| Published pipelines versus their own references in those three states                                | 105       |
| Fair-SMOTE versus matched ERM in those three states: response / utility and group outcomes           | 21 / 18   |
| Controlled learning and matched architectural fit                                                    | 21        |
| INTERFAIR versus removal and additive models: three tasks, two references, nine metrics              | 54        |
| Label-cohort AUROC differences: flagged records / unflagged records                                  | 10 / 10   |

Table S8. Families for joint groups and supporting experiments. Response and utility/outcome columns are separate Holm families. A row with only a response count evaluates edited inputs without a new natural-record utility comparison.

| Scientific comparison                                           | Response           | Utility / outcomes |
|-----------------------------------------------------------------|--------------------|--------------------|
| Joint groups: INTERFAIR versus ERM                              | 10                 | 12                 |
| Joint groups: INTERFAIR versus comparison methods               | 40                 | 44                 |
| Joint groups: published methods versus ERM                      | 30                 | 33                 |
| Transfer: INTERFAIR versus ERM                                  | 40                 | 24                 |
| Transfer: INTERFAIR versus comparison methods                   | 244                | 154                |
| Transfer: published methods versus ERM                          | 204                | 130                |
| Component ablations                                             | 170                | 102                |
| Target allocation: all-relation repair versus ERM               | 30                 | 18                 |
| Target allocation: competing allocations and controls           | 300                | 180                |
| Training-edit policy                                            | 50                 | 30                 |
| Training-context diversity                                      | 60                 | 36                 |
| Maintenance: effect of task-only update                         | 10                 | 6                  |
| Maintenance: recovery after update                              | 20                 | 12                 |
| Maintenance: retention relative to the original INTERFAIR model | 10                 | 6                  |
| Linked income increments                                        | 248 response tests |                    |
| Additional edit values                                          | 26 response tests  |                    |

Paired confidence intervals are unadjusted 95% intervals. Neither an equal rank, a nonsignificant test, nor overlapping intervals establishes that two methods are equivalent.

Audit records participate in neither gradients nor checkpoint selection. Repeated seeds quantify optimization variation on these partitions; they are not ten independent samples of the underlying population. Two-seed DICE and residual-network results in Section S9 are descriptive. Section S7 gives paired intervals for the principal INTERFAIR–ERM changes in utility, group fairness, and decisions (Table S17).

Table S9. What different fairness checks report for two models with the same protected-group offset at \$24k. Model B is the ERM model in the main paper’s running example; Model A replaces its race-dependent interaction with a constant offset equal to Model B’s \$24k gap (−0.384 log-odds). Values are log-odds; score-based attribute-flip checks use tolerance .05.

| Check                                                | Model A: offset only     | Model B: offset and interaction | Distinguishes A from B?              |
|------------------------------------------------------|--------------------------|---------------------------------|--------------------------------------|
| Group metrics (DP, EO [4], AOD)                      | require population rates | require population rates        | not determined by these four outputs |
| Score-based attribute-flip test at \$24k [3, 10, 13] | gap −0.38: fails         | gap −0.38: fails                | no                                   |
| Score-based attribute-flip test at \$34k             | gap −0.38: fails         | gap −0.56: fails                | only by comparing the two profiles   |
| Monotonicity test of the raise [11]                  | +0.11, +0.11: passes     | +0.11, −0.07: fails             | only when a response changes sign    |
| Higher-order rule test                               | $R = 0$ : passes         | $R = -0.18$ : fails             | yes                                  |

**S5 RQ1 (Testing): Attribute-flip checks, reporting codes, and learning controls**

This section supports the paper’s RQ1: whether the generated tests reveal interaction bugs that group metrics and attribute-flip tests miss.

**S5.1 What the fairness checks distinguish**

Table S9 compares the rule test with group metrics, attribute-flip checks, and monotonicity on the main paper’s running example. The two models have the same protected-group gap at the initial income; only one changes that gap when income increases.

**S5.2 Attribute-flip checks on the same executions**

Every test’s four executions also give the first-order relation that attribute-flip testers check: the protected gap at the profile,  $G(x)$ , and at the edited profile,  $G(T(x))$ . A decision-based check flags a test when the decision changes under the flip; a score-based check flags it when  $|G| > .05$ . Table S10 lists, for every model of the paper’s main tables, the percentage of tests that fail equality and the percentages that the two flip checks flag at the profile. Decision-based flags are rare and fall under several mitigations while equality failures stay; score-based flags cover nearly every test before repair and many after it, because they respond to the offset rather than to the rule. Of the 60 published-method results, 38 have fewer decision flips than ERM; in 30 of these, equality failures stay within 5 points of ERM’s or rise.

**S5.3 The linked income edit and the reporting codes**

The linked +\$10k edit applies to 722 of the 2,048 held-out Ohio applicants, 675 of whom pass the support screen, and recomputing their DTI at fixed debt gives a value HMDA never reports for 97.6% of the 722. We evaluated the ten ERM models of the paper’s main tables on the same 675 tests with the recomputed DTI left continuous. Mapping it to the reporting code changes the equality verdict on 52.7 tests per model on average (7.8%), the direction verdict on 17.7 (2.6%), and each residual by .013 on average, although the aggregate failure counts differ little (413 against 406 equality failures in the first seed).

Table S10. Equality failures and attribute-flip flags on the same tests (ten-seed means; percentages of all run tests, pooled over features). Columns: equality failures, decision flips, and score flips. Flips are evaluated at the unedited profile; the paper’s test-suite table also counts flips at the edited profile.

| Model                    | Eq.  | Dec. | Score | Model                   | Eq.  | Dec. | Score |
|--------------------------|------|------|-------|-------------------------|------|------|-------|
| <i>HMDA Ohio / MLP</i>   |      |      |       | <i>Credit / MLP</i>     |      |      |       |
| ERM                      | 80.4 | 3.0  | 98.3  | ERM                     | 24.4 | 1.4  | 87.3  |
| LTDD                     | 87.8 | 3.9  | 96.8  | LTDD                    | 7.2  | 0.6  | 45.2  |
| CoT-Phi                  | 85.4 | 2.5  | 97.8  | CoT-Phi                 | 29.2 | 0.9  | 62.8  |
| DRAlign                  | 79.5 | 2.7  | 92.1  | DRAlign                 | 24.8 | 1.4  | 75.4  |
| Fair-SMOTE               | 86.3 | 9.5  | 96.2  | Fair-SMOTE              | 14.6 | 1.5  | 48.2  |
| Reweighing               | 80.0 | 1.4  | 87.3  | Reweighing              | 25.4 | 0.8  | 59.3  |
| HIFI                     | 79.5 | 3.6  | 97.7  | HIFI                    | 29.4 | 1.4  | 82.5  |
| INTERFAIR                | 0.1  | 0.1  | 5.5   | INTERFAIR               | 0.1  | 0.3  | 22.9  |
| <i>HMDA Maryland</i>     |      |      |       | <i>ACSIIncome / MLP</i> |      |      |       |
| ERM                      | 82.1 | 4.7  | 97.3  | ERM                     | 63.9 | 9.5  | 97.6  |
| LTDD                     | 88.8 | 5.1  | 93.6  | LTDD                    | 52.0 | 3.9  | 88.7  |
| CoT-Phi                  | 85.2 | 3.4  | 94.3  | CoT-Phi                 | 64.8 | 2.6  | 83.1  |
| DRAlign                  | 84.3 | 5.3  | 93.1  | DRAlign                 | 80.2 | 5.9  | 91.3  |
| Fair-SMOTE               | 89.9 | 8.7  | 93.9  | Fair-SMOTE              | 64.0 | 3.0  | 89.0  |
| Reweighing               | 82.2 | 2.9  | 89.3  | Reweighing              | 70.9 | 2.7  | 86.7  |
| HIFI                     | 82.0 | 4.3  | 97.3  | HIFI                    | 68.4 | 7.1  | 96.1  |
| INTERFAIR                | 0.0  | 0.1  | 2.8   | INTERFAIR               | 0.1  | 0.3  | 5.4   |
| <i>HMDA Virginia</i>     |      |      |       | <i>Employment: sex</i>  |      |      |       |
| ERM                      | 80.5 | 3.5  | 97.8  | ERM                     | 82.3 | 6.8  | 95.8  |
| LTDD                     | 85.4 | 3.1  | 89.9  | LTDD                    | 74.1 | 7.2  | 97.4  |
| CoT-Phi                  | 78.9 | 1.9  | 93.0  | CoT-Phi                 | 85.0 | 5.9  | 95.0  |
| DRAlign                  | 84.2 | 4.6  | 90.2  | DRAlign                 | 78.2 | 3.8  | 94.1  |
| Fair-SMOTE               | 89.1 | 8.1  | 92.6  | Fair-SMOTE              | 82.4 | 9.2  | 95.4  |
| Reweighing               | 81.4 | 2.3  | 87.5  | Reweighing              | 80.1 | 5.5  | 94.4  |
| HIFI                     | 82.0 | 3.5  | 96.9  | HIFI                    | 81.1 | 7.9  | 94.8  |
| INTERFAIR                | 0.5  | 0.1  | 27.0  | INTERFAIR               | 4.6  | 3.6  | 89.1  |
| <i>HMDA Pennsylvania</i> |      |      |       | <i>Employment: age</i>  |      |      |       |
| ERM                      | 79.2 | 5.1  | 98.7  | ERM                     | 77.5 | 14.5 | 99.6  |
| LTDD                     | 89.9 | 9.1  | 97.4  | LTDD                    | 86.5 | 15.6 | 99.5  |
| CoT-Phi                  | 85.5 | 5.3  | 98.5  | CoT-Phi                 | 87.2 | 15.3 | 97.5  |
| DRAlign                  | 80.9 | 4.4  | 89.8  | DRAlign                 | 77.4 | 6.3  | 94.7  |
| Fair-SMOTE               | 82.9 | 10.2 | 97.4  | Fair-SMOTE              | 81.1 | 13.1 | 97.2  |
| Reweighing               | 77.6 | 2.4  | 90.5  | Reweighing              | 78.1 | 12.3 | 96.8  |
| HIFI                     | 80.9 | 5.0  | 98.0  | HIFI                    | 85.9 | 16.0 | 98.9  |
| INTERFAIR                | 0.1  | 0.1  | 6.0   | INTERFAIR               | 1.1  | 0.2  | 17.3  |
| <i>HMDA Ohio / FT</i>    |      |      |       | <i>ACSIIncome / FT</i>  |      |      |       |
| ERM                      | 72.6 | 2.5  | 98.3  | ERM                     | 76.5 | 9.2  | 90.0  |
| LTDD                     | 83.8 | 2.5  | 95.0  | LTDD                    | 55.4 | 4.6  | 76.6  |
| CoT-Phi                  | 81.8 | 2.1  | 95.1  | CoT-Phi                 | 65.7 | 3.5  | 76.5  |
| DRAlign                  | 0.0  | 0.2  | 1.0   | DRAlign                 | 3.7  | 1.2  | 21.8  |
| Fair-SMOTE               | 86.5 | 9.4  | 95.6  | Fair-SMOTE              | 69.4 | 4.7  | 80.7  |
| Reweighing               | 42.8 | 0.9  | 69.1  | Reweighing              | 72.5 | 4.3  | 79.7  |
| HIFI                     | 69.9 | 2.6  | 97.3  | HIFI                    | 73.8 | 6.3  | 87.0  |
| INTERFAIR                | 0.0  | 0.1  | 13.6  | INTERFAIR               | 0.0  | 0.1  | 0.1   |

S5.4 Where the violations arise: learning controls

The controlled-label experiment asks whether a learner can introduce a protected-dependent rule when the score that generated its labels has no such interaction. We retain the real input profiles

Table S11. Mean residuals in the learning controls (ten seeds). Generated labels come from a score with exactly zero protected-by-feature interaction. All columns use the matched control protocol.

| Task / MLP | Real labels | Protected column permuted | Generated labels |
|------------|-------------|---------------------------|------------------|
| HMDA Ohio  | .1882       | .1834                     | .1633            |
| Credit     | .0328       | .0330                     | .0293            |
| ACSIIncome | .1312       | .0561                     | .0648            |

and their correlations, fit a generating score  $z^*(a, x) = g(x) + b_a$ , and draw binary labels from  $\sigma(z^*)$ . The shared function  $g$  can use arbitrary ordinary-feature interactions; the protected input contributes only a constant offset. Its protected-by-edit logit residual is therefore exactly zero. We then fit an unrestricted MLP to the generated labels and audit its responses.

Three independent label draws are averaged within each of ten optimizer seeds. A real-label arm uses the same fitting protocol. Table S11 shows that the resulting learned residual is 87% of its real-label counterpart on HMDA, 89% on Credit, and 49% on ACSIIncome. Thus, ordinary supervised learning need not preserve an exact equal-response property of the generating score.

A second control permutes the protected column during training while retaining the other inputs and real labels; validation and audit use the original protected values. Its residual is 97%, 101%, and 43% of the respective real-label residual. On HMDA and Credit, training therefore produces most of the interaction even when the protected column carries no information about the label.

## S6 RQ2 (Localization): Stability and repair from failing tests

This section supports the paper’s RQ2: whether failing tests localize the bugs to features and point to what to repair. It gives the stability of the per-feature localization, the discovery and allocation protocol of the repair study, its two control arms, and the replay of the saved tests.

### S6.1 Stability of the localization

Table S12 compares the per-feature mean residuals of each pair of training runs. A concentrated violation, such as the tract-share violation that LTDD and CoT-Phi introduce, is ranked first in nearly every run; ERM’s violations, spread over all features, give a less stable ranking.

### S6.2 Discovery, allocation, and matched controls

A discovery audit uses 1,024 training records to identify failing relations. Those records, or their complete households in ACSIIncome, are excluded from the contexts later used for direct steering. Natural-record population penalties still use the training partition. All repair arms train anew from paired random initialization for 40 epochs; they do not continue the discovered ERM’s weights.

A relation is selected when a test displays a consequential equality failure or a wrong-direction response. Consequential equality failures have  $|R| > .05$  and either opposite probability responses beyond .005 or a decision discrepancy removed by eliminating the local interaction. For the latter check, write the four logits as  $(z_{00}, z_{01}, z_{10}, z_{11})$  and subtract  $(R/4)(1, -1, -1, 1)$ . This centered additive projection removes the residual while preserving the four-score mean and the average protected and edit effects. We require both original and projected probabilities to be farther than .005 from the decision threshold. The projection prioritizes tests; it is not used as an output repair.

Every relation containing such a failing test is eligible; there is no fixed quota of features. Repair on failing relations applies the rule penalties throughout the selected relations; repair on failing contexts further restricts them to the failing contexts identified in the separate steering pool. The

Table S12. Stability of the per-feature localization across the ten training runs (HMDA, MLP). Rank correlation is Spearman’s between the per-feature mean  $|R|$  of two runs: median over the 45 pairs, with the minimum in parentheses. The top feature has the largest ten-run mean residual; the last column counts the runs in which it also ranks first.

| State        | Model   | Rank correlation | Top feature          | Runs  |
|--------------|---------|------------------|----------------------|-------|
| Ohio         | ERM     | 0.67 (0.35)      | Loan type            | 3/10  |
| Ohio         | LTDD    | 0.84 (0.72)      | Tract minority share | 9/10  |
| Ohio         | CoT-Phi | 0.81 (0.55)      | Tract minority share | 10/10 |
| Maryland     | ERM     | 0.89 (0.73)      | Loan-to-value        | 6/10  |
| Maryland     | LTDD    | 0.88 (0.68)      | Loan type            | 7/10  |
| Maryland     | CoT-Phi | 0.85 (0.67)      | Tract minority share | 10/10 |
| Virginia     | ERM     | 0.65 (0.39)      | Preapproval          | 3/10  |
| Virginia     | LTDD    | 0.85 (0.62)      | Tract minority share | 9/10  |
| Virginia     | CoT-Phi | 0.64 (0.10)      | Loan type            | 8/10  |
| Pennsylvania | ERM     | 0.76 (0.46)      | Loan-to-value        | 7/10  |
| Pennsylvania | LTDD    | 0.90 (0.74)      | Tract minority share | 8/10  |
| Pennsylvania | CoT-Phi | 0.82 (0.52)      | Tract minority share | 10/10 |

outcome terms retain all features in both cases, so the study compares allocations of the rule penalties.

All allocation arms use identical task batches and optimizer updates. Separate query streams supply population records, population effects, ordinary direct relations, signed relations, and an augmentation slot. Unused slots are still queried without adding a loss, giving 4,096 regularizer rows per update in every arm. This matches compute rather than granting a narrower allocation fewer optimization steps.

The task-only control has no fairness training penalty but uses INTERFAIR’s validation selector. Across the 30 paired runs, its 240 candidate checkpoints have identical measurements to ERM’s corresponding candidates. Selection changes five Credit seeds and one HMDA seed; it never changes ACSIncome. Table S13 separates that selection effect from learned repair.

The augmentation arm retrain on the failing records and their protected-flipped copies, each labeled with the original record’s label, under the matched query and fitting budget; it tests labeled examples against relational supervision. It follows the test-then-retrain strategy of fairness testers such as AEQUITAS [13], without AEQUITAS’s adaptive enlargement and acceptance procedure. On ACSIncome it leaves  $V = 64.55\%$ , against  $.80\%$  for repair on failing relations.

### S6.3 Replaying tests and preserving behavior

Table S14 replays the saved tests after repair on failing relations and separates failures that disappear from new failures on formerly passing tests.

A selected failure is resolved when its triggering behavioral event disappears, not merely when its residual shrinks. Counts pool tests within each seed, unlike the feature-macro averages  $V$  and  $W$ . The matched task-only control is necessary because checkpoint changes can alter which tests fail even without a fairness penalty.

Context restriction has a measurable preservation trade-off on ACSIncome. Repair on all relations, on failing relations, and on failing contexts changes natural probabilities by means of 6.00, 6.06, and 5.74 percentage points. Their mean response changes are 3.88, 3.85, and 3.32 points, while equality violations are  $.12\%$ ,  $.80\%$ , and  $3.79\%$ . Context restriction therefore preserves somewhat more behavior but satisfies fewer equality tests. These drift comparisons are descriptive.

Table S13. Checkpoint selection and context restriction in the targeted-repair experiment. All arms start from paired random weights and receive the same number of updates and queries. Ten-seed means; rates are percentages. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Training                       | AUROC↑                    | AOD↓                      | $\bar{R}$ ↓               | V ↓                      | D ↓                      | W ↓                      |
|--------------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| <i>HMDA Ohio / MLP</i>         |                           |                           |                           |                          |                          |                          |
| ERM                            | <b>0.850</b> <sup>1</sup> | 0.104 <sup>5</sup>        | 0.200 <sup>5</sup>        | 80.73 <sup>7</sup>       | 5.34 <sup>6</sup>        | 26.13 <sup>5</sup>       |
| Task loss, INTERFAIR selection | <b>0.850</b> <sup>1</sup> | 0.096 <sup>4</sup>        | 0.191 <sup>5</sup>        | 79.47 <sup>6</sup>       | 5.20 <sup>6</sup>        | 25.63 <sup>5</sup>       |
| INTERFAIR, failing relations   | 0.844 <sup>4</sup>        | <b>0.012</b> <sup>1</sup> | <b>0.007</b> <sup>1</sup> | <b>0.08</b> <sup>2</sup> | <b>0.13</b> <sup>2</sup> | 0.76 <sup>2</sup>        |
| INTERFAIR, failing contexts    | 0.846 <sup>2</sup>        | <b>0.011</b> <sup>1</sup> | 0.008 <sup>2</sup>        | 0.14 <sup>3</sup>        | 0.14 <sup>3</sup>        | <b>0.55</b> <sup>1</sup> |
| <i>Credit / MLP</i>            |                           |                           |                           |                          |                          |                          |
| ERM                            | 0.785 <sup>2</sup>        | 0.025 <sup>5</sup>        | 0.040 <sup>6</sup>        | 24.43 <sup>6</sup>       | 1.90 <sup>5</sup>        | 6.62 <sup>6</sup>        |
| Task loss, INTERFAIR selection | 0.784 <sup>3</sup>        | 0.021 <sup>4</sup>        | 0.024 <sup>5</sup>        | 12.16 <sup>5</sup>       | 1.38 <sup>4</sup>        | 3.22 <sup>5</sup>        |
| INTERFAIR, failing relations   | <b>0.786</b> <sup>1</sup> | <b>0.011</b> <sup>1</sup> | <b>0.010</b> <sup>2</sup> | <b>0.39</b> <sup>2</sup> | <b>0.51</b> <sup>1</sup> | 0.46 <sup>3</sup>        |
| INTERFAIR, failing contexts    | 0.784 <sup>3</sup>        | 0.014 <sup>2</sup>        | 0.012 <sup>3</sup>        | 0.84 <sup>3</sup>        | 0.76 <sup>2</sup>        | <b>0.24</b> <sup>1</sup> |
| <i>ACSIIncome / MLP</i>        |                           |                           |                           |                          |                          |                          |
| ERM                            | <b>0.905</b> <sup>1</sup> | 0.070 <sup>4</sup>        | 0.107 <sup>5</sup>        | 63.90 <sup>5</sup>       | 13.82 <sup>6</sup>       | 0.13 <sup>4</sup>        |
| Task loss, INTERFAIR selection | <b>0.905</b> <sup>1</sup> | 0.070 <sup>4</sup>        | 0.107 <sup>5</sup>        | 63.90 <sup>5</sup>       | 13.82 <sup>6</sup>       | 0.13 <sup>4</sup>        |
| INTERFAIR, failing relations   | 0.898 <sup>4</sup>        | <b>0.014</b> <sup>1</sup> | <b>0.011</b> <sup>2</sup> | <b>0.80</b> <sup>2</sup> | <b>0.66</b> <sup>2</sup> | 0.11 <sup>4</sup>        |
| INTERFAIR, failing contexts    | 0.899 <sup>3</sup>        | <b>0.014</b> <sup>1</sup> | 0.016 <sup>3</sup>        | 3.79 <sup>3</sup>        | 1.20 <sup>3</sup>        | <b>0.04</b> <sup>2</sup> |

Table S14. Replay after repair on failing relations. Ten-seed mean percentages. Resolution is conditional on the original failure set; new failures are conditional on the original passing set.

| Task       | Resolved          |                   | Newly failing |                |
|------------|-------------------|-------------------|---------------|----------------|
|            | Selected failures | Equality failures | Equality      | Other behavior |
| HMDA Ohio  | 98.2              | 99.9              | .1            | .1             |
| Credit     | 92.1              | 97.9              | .3            | .5             |
| ACSIIncome | 98.3              | 99.1              | .7            | .5             |

## S7 RQ3 (Repair): Variation, cost, and component contributions

This section supports the paper’s RQ3: whether interaction steering repairs the rule, at what cost, and what each of its terms contributes.

### S7.1 FT-Transformer comparison

Table S15 completes the main paper’s comparison with the two FT-Transformer settings. It uses the same metrics, ten-seed aggregation, and Scott–Knott ESD conventions as the MLP tables.

On HMDA with FT-Transformer, DRAlign’s smaller residual (.0017 versus .0051 for INTERFAIR) accompanies strong response contraction: the mean magnitude of the probability response shared by the two groups falls from .0705 under ERM to .0072, against .0354 under INTERFAIR (Table S18), while AUROC falls from .857 to .751 (INTERFAIR: .847).

### S7.2 Variation across runs and paired intervals

The main comparisons show the mean outcome of every evaluated method. Table S16 adds standard deviations for the central ERM–INTERFAIR contrast across all ten settings. In each setting, the

Table S15. FT-Transformer on HMDA Ohio (race) and ACSIncome (sex). Ten-seed held-out means with Scott-Knott ESD groups; bold marks the best displayed group.  $V$ ,  $D$ , and  $W$  are percentages. For ACSIncome  $W$ , ESD grouping is undefined because several methods have zero variance; bold marks the lowest observed rate.

| Method                            | AUROC↑                    | Acc.↑                     | F1↑                       | AOD↓                      | DP↓                       | EO↓                       | $\bar{R}$ ↓               | V ↓                      | D ↓                      | W ↓                      |
|-----------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| utility and group outcomes        |                           |                           |                           |                           |                           |                           | rule-level behavior       |                          |                          |                          |
| <i>HMDA Ohio / FT-Transformer</i> |                           |                           |                           |                           |                           |                           |                           |                          |                          |                          |
| ERM                               | <b>0.857</b> <sup>1</sup> | <b>0.873</b> <sup>1</sup> | 0.925 <sup>2</sup>        | 0.096 <sup>6</sup>        | 0.142 <sup>7</sup>        | 0.148 <sup>6</sup>        | 0.202 <sup>6</sup>        | 73.34 <sup>7</sup>       | 4.46 <sup>6</sup>        | 25.61 <sup>7</sup>       |
| LTDD                              | 0.855 <sup>3</sup>        | 0.873 <sup>2</sup>        | 0.925 <sup>3</sup>        | 0.032 <sup>3</sup>        | 0.091 <sup>4</sup>        | 0.049 <sup>3</sup>        | 0.295 <sup>7</sup>        | 84.05 <sup>9</sup>       | 4.35 <sup>6</sup>        | 25.28 <sup>7</sup>       |
| CoT-Phi                           | 0.854 <sup>4</sup>        | 0.872 <sup>3</sup>        | 0.925 <sup>4</sup>        | 0.055 <sup>4</sup>        | 0.108 <sup>5</sup>        | 0.085 <sup>4</sup>        | 0.286 <sup>7</sup>        | 81.93 <sup>8</sup>       | 3.66 <sup>5</sup>        | 20.08 <sup>5</sup>       |
| DRAlign                           | 0.751 <sup>9</sup>        | 0.745 <sup>9</sup>        | 0.802 <sup>10</sup>       | <b>0.006</b> <sup>1</sup> | <b>0.011</b> <sup>1</sup> | <b>0.007</b> <sup>1</sup> | <b>0.002</b> <sup>1</sup> | <b>0.00</b> <sup>1</sup> | 0.37 <sup>2</sup>        | 5.99 <sup>2</sup>        |
| Fair-SMOTE                        | 0.835 <sup>8</sup>        | 0.830 <sup>8</sup>        | 0.893 <sup>9</sup>        | 0.076 <sup>5</sup>        | 0.146 <sup>7</sup>        | 0.090 <sup>4</sup>        | 0.403 <sup>8</sup>        | 86.60 <sup>10</sup>      | 11.11 <sup>9</sup>       | 40.35 <sup>8</sup>       |
| Reweighting                       | 0.855 <sup>3</sup>        | 0.871 <sup>4</sup>        | 0.925 <sup>5</sup>        | 0.013 <sup>2</sup>        | 0.067 <sup>3</sup>        | 0.023 <sup>2</sup>        | 0.087 <sup>3</sup>        | 43.89 <sup>3</sup>       | 2.23 <sup>3</sup>        | 17.74 <sup>4</sup>       |
| HIFI                              | 0.856 <sup>2</sup>        | <b>0.873</b> <sup>1</sup> | <b>0.925</b> <sup>1</sup> | 0.098 <sup>6</sup>        | 0.144 <sup>7</sup>        | 0.150 <sup>6</sup>        | 0.186 <sup>5</sup>        | 70.59 <sup>6</sup>       | 4.66 <sup>7</sup>        | 22.81 <sup>6</sup>       |
| <b>INTERFAIR</b>                  | 0.847 <sup>6</sup>        | 0.862 <sup>6</sup>        | 0.921 <sup>7</sup>        | 0.013 <sup>2</sup>        | 0.039 <sup>2</sup>        | 0.024 <sup>2</sup>        | 0.005 <sup>2</sup>        | 0.02 <sup>2</sup>        | <b>0.11</b> <sup>1</sup> | <b>0.19</b> <sup>1</sup> |
| <i>ACSIncome / FT-Transformer</i> |                           |                           |                           |                           |                           |                           |                           |                          |                          |                          |
| ERM                               | <b>0.907</b> <sup>1</sup> | <b>0.822</b> <sup>1</sup> | <b>0.819</b> <sup>1</sup> | 0.060 <sup>7</sup>        | 0.143 <sup>8</sup>        | 0.068 <sup>7</sup>        | 0.228 <sup>7</sup>        | 76.42 <sup>8</sup>       | 13.11 <sup>8</sup>       | <b>0.00</b> <sup>1</sup> |
| LTDD                              | 0.905 <sup>3</sup>        | 0.821 <sup>2</sup>        | 0.816 <sup>4</sup>        | 0.019 <sup>4</sup>        | 0.101 <sup>5</sup>        | 0.026 <sup>4</sup>        | 0.147 <sup>4</sup>        | 55.34 <sup>3</sup>       | 8.58 <sup>5</sup>        | 0.00                     |
| CoT-Phi                           | 0.904 <sup>4</sup>        | 0.820 <sup>3</sup>        | 0.815 <sup>4</sup>        | 0.014 <sup>3</sup>        | 0.093 <sup>4</sup>        | 0.022 <sup>3</sup>        | 0.136 <sup>3</sup>        | 65.63 <sup>4</sup>       | 6.43 <sup>3</sup>        | 0.00                     |
| DRAlign                           | 0.831 <sup>8</sup>        | 0.609 <sup>8</sup>        | 0.432 <sup>8</sup>        | 0.022 <sup>5</sup>        | <b>0.043</b> <sup>1</sup> | 0.036 <sup>5</sup>        | 0.010 <sup>2</sup>        | 3.68 <sup>2</sup>        | 1.73 <sup>2</sup>        | 4.30                     |
| Fair-SMOTE                        | 0.905 <sup>3</sup>        | 0.819 <sup>4</sup>        | 0.818 <sup>2</sup>        | 0.015 <sup>3</sup>        | 0.092 <sup>4</sup>        | 0.020 <sup>2</sup>        | 0.143 <sup>4</sup>        | 69.42 <sup>5</sup>       | 7.58 <sup>4</sup>        | <b>0.00</b> <sup>1</sup> |
| Reweighting                       | 0.905 <sup>2</sup>        | 0.820 <sup>3</sup>        | 0.815 <sup>4</sup>        | 0.012 <sup>2</sup>        | 0.088 <sup>3</sup>        | <b>0.017</b> <sup>1</sup> | 0.168 <sup>5</sup>        | 72.42 <sup>6</sup>       | 7.45 <sup>4</sup>        | 0.01                     |
| HIFI                              | <b>0.907</b> <sup>1</sup> | 0.821 <sup>2</sup>        | 0.817 <sup>3</sup>        | 0.048 <sup>6</sup>        | 0.130 <sup>6</sup>        | 0.056 <sup>6</sup>        | 0.193 <sup>6</sup>        | 73.73 <sup>7</sup>       | 9.57 <sup>6</sup>        | <b>0.00</b> <sup>1</sup> |
| <b>INTERFAIR</b>                  | 0.901 <sup>5</sup>        | 0.815 <sup>5</sup>        | 0.812 <sup>5</sup>        | <b>0.010</b> <sup>1</sup> | 0.076 <sup>2</sup>        | <b>0.016</b> <sup>1</sup> | <b>0.005</b> <sup>1</sup> | <b>0.01</b> <sup>1</sup> | <b>0.30</b> <sup>1</sup> | <b>0.00</b> <sup>1</sup> |

two methods are queried on identical held-out tests. Residual and decision-response disagreement decrease in all 100 paired runs. The highest remaining equality-violation rate is 4.4%, on ACSEmployment by sex.

On held-out records, AOD improves in 99 of the 100 paired runs, DP in 98, and EO in 94, and every setting’s mean gap improves; the per-seed export lists each run.

Table S17 gives the paired INTERFAIR–ERM changes in AUROC, AOD, and  $D$  with 95% intervals; every AOD and  $D$  change and nine of the ten AUROC changes survive Holm correction.

### S7.3 Response changes beyond score contraction

Multiplying all logits by a positive constant shrinks logit residuals without changing prediction rankings, response signs, or decisions at threshold .5. The observed reductions in decision disagreement and in zero-margin wrong-direction responses go beyond that transformation. Table S18 reports those signs alongside response magnitude and Brier score, the mean squared probability error against the recorded labels.

INTERFAIR retains 50–101% of ERM’s mean common probability response, and its Brier score is higher in every setting; the response signs and decision measures show changes beyond uniform score scaling.

### S7.4 Locating the utility difference with situation testing

Fairness methods are usually evaluated against recorded outcomes, which can themselves differ between comparable members of different groups. Situation testing [9] asks where those differences occur. We apply it to labels solely to characterize the utility cost; it changes neither training labels nor model predictions.

Table S16. Variation across training runs on the main audit banks. Entries are means with sample standard deviations over ten seeds. Rate columns are percentages. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Method                         | $\bar{R} \downarrow$                       | $V \downarrow$                           | $D \downarrow$                           | $W \downarrow$                           |
|--------------------------------|--------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| <i>HMDA Ohio / MLP</i>         |                                            |                                          |                                          |                                          |
| ERM                            | 0.200 <sup>4</sup> <sub>±0.052</sub>       | 80.73 <sup>3</sup> <sub>±4.29</sub>      | 5.34 <sup>5</sup> <sub>±1.09</sub>       | 26.13 <sup>5</sup> <sub>±3.48</sub>      |
| <b>INTERFAIR</b>               | <b>0.008<sup>1</sup></b> <sub>±0.003</sub> | <b>0.10<sup>1</sup></b> <sub>±0.13</sub> | <b>0.13<sup>1</sup></b> <sub>±0.06</sub> | <b>0.68<sup>1</sup></b> <sub>±0.42</sub> |
| <i>HMDA Maryland / MLP</i>     |                                            |                                          |                                          |                                          |
| ERM                            | 0.225 <sup>3</sup> <sub>±0.044</sub>       | 82.44 <sup>2</sup> <sub>±3.72</sub>      | 6.53 <sup>5</sup> <sub>±1.35</sub>       | 30.01 <sup>4</sup> <sub>±2.98</sub>      |
| <b>INTERFAIR</b>               | <b>0.006<sup>1</sup></b> <sub>±0.001</sub> | <b>0.03<sup>1</sup></b> <sub>±0.04</sub> | <b>0.15<sup>1</sup></b> <sub>±0.03</sub> | <b>0.73<sup>1</sup></b> <sub>±0.46</sub> |
| <i>HMDA Virginia / MLP</i>     |                                            |                                          |                                          |                                          |
| ERM                            | 0.191 <sup>3</sup> <sub>±0.038</sub>       | 80.71 <sup>3</sup> <sub>±3.62</sub>      | 4.73 <sup>4</sup> <sub>±0.94</sub>       | 27.32 <sup>4</sup> <sub>±3.09</sub>      |
| <b>INTERFAIR</b>               | <b>0.012<sup>1</sup></b> <sub>±0.002</sub> | <b>0.53<sup>1</sup></b> <sub>±0.63</sub> | <b>0.27<sup>1</sup></b> <sub>±0.10</sub> | <b>0.50<sup>1</sup></b> <sub>±0.29</sub> |
| <i>HMDA Pennsylvania / MLP</i> |                                            |                                          |                                          |                                          |
| ERM                            | 0.187 <sup>3</sup> <sub>±0.037</sub>       | 79.43 <sup>3</sup> <sub>±4.09</sub>      | 7.12 <sup>4</sup> <sub>±1.48</sub>       | 29.36 <sup>5</sup> <sub>±3.21</sub>      |
| <b>INTERFAIR</b>               | <b>0.008<sup>1</sup></b> <sub>±0.002</sub> | <b>0.06<sup>1</sup></b> <sub>±0.08</sub> | <b>0.17<sup>1</sup></b> <sub>±0.10</sub> | <b>0.88<sup>1</sup></b> <sub>±0.46</sub> |
| <i>Credit / MLP</i>            |                                            |                                          |                                          |                                          |
| ERM                            | 0.040 <sup>4</sup> <sub>±0.028</sub>       | 24.43 <sup>4</sup> <sub>±20.35</sub>     | 1.90 <sup>5</sup> <sub>±0.93</sub>       | 6.62 <sup>2</sup> <sub>±5.69</sub>       |
| <b>INTERFAIR</b>               | <b>0.010<sup>1</sup></b> <sub>±0.001</sub> | <b>0.12<sup>1</sup></b> <sub>±0.05</sub> | <b>0.54<sup>1</sup></b> <sub>±0.25</sub> | <b>0.36<sup>1</sup></b> <sub>±0.18</sub> |
| <i>ACSIncome / MLP</i>         |                                            |                                          |                                          |                                          |
| ERM                            | 0.107 <sup>3</sup> <sub>±0.017</sub>       | 63.90 <sup>3</sup> <sub>±6.93</sub>      | 13.82 <sup>8</sup> <sub>±0.93</sub>      | 0.13 <sup>3</sup> <sub>±0.08</sub>       |
| <b>INTERFAIR</b>               | <b>0.010<sup>1</sup></b> <sub>±0.001</sub> | <b>0.14<sup>1</sup></b> <sub>±0.17</sub> | <b>0.49<sup>1</sup></b> <sub>±0.13</sub> | <b>0.04<sup>1</sup></b> <sub>±0.05</sub> |
| <i>Employment: sex / MLP</i>   |                                            |                                          |                                          |                                          |
| ERM                            | 0.194 <sup>5</sup> <sub>±0.031</sub>       | 82.58 <sup>6</sup> <sub>±2.42</sub>      | 7.01 <sup>4</sup> <sub>±1.04</sub>       | 8.09 <sup>4</sup> <sub>±5.85</sub>       |
| <b>INTERFAIR</b>               | <b>0.018<sup>1</sup></b> <sub>±0.003</sub> | <b>4.44<sup>1</sup></b> <sub>±2.33</sub> | <b>2.04<sup>1</sup></b> <sub>±0.69</sub> | <b>0.18<sup>1</sup></b> <sub>±0.21</sub> |
| <i>Employment: age / MLP</i>   |                                            |                                          |                                          |                                          |
| ERM                            | 0.142 <sup>2</sup> <sub>±0.033</sub>       | 76.86 <sup>2</sup> <sub>±4.32</sub>      | 15.54 <sup>3</sup> <sub>±2.60</sub>      | 2.80 <sup>2</sup> <sub>±4.05</sub>       |
| <b>INTERFAIR</b>               | <b>0.012<sup>1</sup></b> <sub>±0.001</sub> | <b>1.06<sup>1</sup></b> <sub>±0.51</sub> | <b>0.46<sup>1</sup></b> <sub>±0.21</sub> | <b>0.07<sup>1</sup></b> <sub>±0.07</sub> |
| <i>HMDA Ohio / FT</i>          |                                            |                                          |                                          |                                          |
| ERM                            | 0.202 <sup>6</sup> <sub>±0.031</sub>       | 73.34 <sup>7</sup> <sub>±3.52</sub>      | 4.46 <sup>6</sup> <sub>±0.94</sub>       | 25.61 <sup>7</sup> <sub>±2.93</sub>      |
| <b>INTERFAIR</b>               | <b>0.005<sup>2</sup></b> <sub>±0.001</sub> | <b>0.02<sup>2</sup></b> <sub>±0.02</sub> | <b>0.11<sup>1</sup></b> <sub>±0.06</sub> | <b>0.19<sup>1</sup></b> <sub>±0.23</sub> |
| <i>ACSIncome / FT</i>          |                                            |                                          |                                          |                                          |
| ERM                            | 0.228 <sup>7</sup> <sub>±0.040</sub>       | 76.42 <sup>8</sup> <sub>±3.39</sub>      | 13.11 <sup>8</sup> <sub>±1.45</sub>      | <b>0.00</b> <sub>±0.00</sub>             |
| <b>INTERFAIR</b>               | <b>0.005<sup>1</sup></b> <sub>±0.001</sub> | <b>0.01<sup>1</sup></b> <sub>±0.02</sub> | <b>0.30<sup>1</sup></b> <sub>±0.10</sub> | <b>0.00</b> <sub>±0.00</sub>             |

For each audit record, we find its ten nearest training records in its own protected group and ten in the other group. Distance uses the frozen encoded inputs with the protected coordinate removed. Let  $p_{\text{same}}$  and  $p_{\text{other}}$  be the favorable-label rates in those neighborhoods. For the audit record's label  $y$ , define

$$S = (2y - 1)(p_{\text{same}} - p_{\text{other}}). \quad (\text{S2})$$

A record is flagged when  $S \geq .2$ : its observed label agrees more with the own-group pattern than with the other group's pattern by at least 20 percentage points. For a recorded denial, the

Table S17. INTERFAIR minus ERM: mean paired changes and 95% confidence intervals, in percentage points. Negative AUROC changes are utility losses; negative AOD and  $D$  changes are improvements.

| Setting              | $\Delta$ AUROC        | $\Delta$ AOD             | $\Delta D$               |
|----------------------|-----------------------|--------------------------|--------------------------|
| Ohio / MLP           | -0.83 [-1.22, -0.44]* | -9.59 [-11.44, -7.74]*   | -5.21 [-6.02, -4.40]*    |
| Maryland / MLP       | -0.71 [-0.90, -0.53]* | -5.63 [-7.11, -4.15]*    | -6.38 [-7.34, -5.42]*    |
| Virginia / MLP       | -1.48 [-1.74, -1.22]* | -2.55 [-3.40, -1.71]*    | -4.46 [-5.09, -3.82]*    |
| Pennsylvania / MLP   | -0.85 [-1.10, -0.59]* | -12.36 [-14.22, -10.51]* | -6.95 [-7.98, -5.92]*    |
| Credit / MLP         | -0.06 [-0.41, +0.28]  | -1.62 [-2.30, -0.95]*    | -1.37 [-2.01, -0.72]*    |
| ACSIIncome / MLP     | -0.79 [-1.18, -0.40]* | -5.65 [-6.31, -4.99]*    | -13.33 [-14.02, -12.64]* |
| Employment sex / MLP | -0.95 [-1.12, -0.78]* | -1.09 [-1.62, -0.56]*    | -4.97 [-5.94, -4.00]*    |
| Employment age / MLP | -1.31 [-1.48, -1.15]* | -3.27 [-4.33, -2.22]*    | -15.09 [-17.02, -13.16]* |
| Ohio / FT            | -0.99 [-1.22, -0.77]* | -8.31 [-9.99, -6.63]*    | -4.36 [-5.03, -3.68]*    |
| ACSIIncome / FT      | -0.57 [-0.75, -0.39]* | -4.98 [-5.64, -4.32]*    | -12.81 [-13.87, -11.76]* |

Intervals are unadjusted paired intervals across ten seeds. \* marks  $p_{\text{Holm}} < .05$  within its comparison family; family definitions appear in Section S4.

Table S18. Response signs, response magnitude, and probability quality.  $W_0$  counts any wrong-direction response without a reporting margin;  $C_P$  is mean common probability-response magnitude (both in percent).

| Setting              | $W_0 \downarrow$ ERM | $W_0 \downarrow$ INTERFAIR | $C_P$ ERM | $C_P$ INTERFAIR | $\Delta$ Brier |
|----------------------|----------------------|----------------------------|-----------|-----------------|----------------|
| Ohio / MLP           | 33.48 <sup>2</sup>   | <b>1.62</b> <sup>1</sup>   | 8.03      | 4.71            | +0.0048        |
| Maryland / MLP       | 39.36 <sup>2</sup>   | <b>1.90</b> <sup>1</sup>   | 8.04      | 4.43            | +0.0033        |
| Virginia / MLP       | 36.70 <sup>2</sup>   | <b>1.71</b> <sup>1</sup>   | 6.78      | 3.88            | +0.0084        |
| Pennsylvania / MLP   | 35.03 <sup>2</sup>   | <b>2.05</b> <sup>1</sup>   | 8.22      | 4.74            | +0.0038        |
| Credit / MLP         | 11.34 <sup>2</sup>   | <b>3.04</b> <sup>1</sup>   | 2.90      | 2.09            | +0.0014        |
| ACSIIncome / MLP     | 0.36 <sup>2</sup>    | <b>0.07</b> <sup>1</sup>   | 11.98     | 11.55           | +0.0055        |
| Employment sex / MLP | 10.54 <sup>2</sup>   | <b>0.61</b> <sup>1</sup>   | 4.71      | 2.72            | +0.0043        |
| Employment age / MLP | 3.92 <sup>2</sup>    | <b>0.28</b> <sup>1</sup>   | 6.52      | 6.59            | +0.0051        |
| Ohio / FT            | 36.34 <sup>2</sup>   | <b>2.48</b> <sup>1</sup>   | 7.05      | 3.54            | +0.0065        |
| ACSIIncome / FT      | 0.68 <sup>2</sup>    | <b>0.08</b> <sup>1</sup>   | 13.61     | 10.65           | +0.0024        |

$C_P$  averages  $|(\Delta p_0 + \Delta p_1)/2|$  over the ordinary edits.  $\Delta$ Brier is INTERFAIR minus ERM; positive values indicate worse squared-probability error. For  $W_0$ , superscripts compare ERM and INTERFAIR within each row; the best Scott–Knott ESD group is bold. Response magnitude has no preferred direction and is not ranked.

other-group neighbors must have been approved more often. This score uses training labels and the audit record’s label, but no model outputs. The same flagged set therefore applies to every method and seed in a setting.

We compute AUROC separately on flagged and unflagged records. Table S19 reports the resulting INTERFAIR–ERM differences: 12–28% of records are flagged, and the difference is more negative on those records. On the unflagged records, INTERFAIR has significantly lower AUROC only in Virginia; it has significantly higher AUROC in Maryland and ACSEmployment by age. The other seven comparisons are inconclusive. These are comparisons of ranking within each subset. Overall AUROC also includes favorable–unfavorable pairs spanning the subsets, so the two subset AUROCs are not an additive decomposition of the overall loss.

**Controls.** The random-subset control takes 20 subsets per seed, each as large as the unflagged set. Their average AUROC change recovers the full-sample change (Table S20). Merely reducing the evaluation size therefore does not explain the smaller gap on unflagged labels.

Table S19. Where INTERFAIR’s AUROC loss lies: INTERFAIR – ERM in AUROC, in percentage points (–0.83 means, e.g., .850 to .842; ten-seed means), on all audit records, on records whose recorded label situation testing flags as group-patterned (similar people of the other protected group more often received the opposite outcome), and on the other records.  $\uparrow/\downarrow$ : significantly higher/lower than ERM (paired  $t$ -tests over seeds, Holm across the ten settings).

|                             | MLP                |                    |                    |                    |        |                    |                    |                    | FT-Transformer     |                    |
|-----------------------------|--------------------|--------------------|--------------------|--------------------|--------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                             | HMDA               |                    |                    |                    |        |                    | Employment         |                    | HMDA               | Income             |
|                             | OH                 | MD                 | VA                 | PA                 | Credit | Income             | sex                | age                |                    |                    |
| Flagged labels (%)          | 19.7               | 16.0               | 15.8               | 21.5               | 12.2   | 20.6               | 22.3               | 28.5               | 19.7               | 20.6               |
| $\Delta$ AUROC, all records | –0.83              | –0.71              | –1.48              | –0.85              | –0.06  | –0.79              | –0.95              | –1.31              | –0.99              | –0.57              |
| flagged records             | –3.92 $\downarrow$ | –6.27 $\downarrow$ | –6.97 $\downarrow$ | –3.71 $\downarrow$ | –0.75  | –3.88 $\downarrow$ | –4.14 $\downarrow$ | –7.42 $\downarrow$ | –3.35 $\downarrow$ | –4.59 $\downarrow$ |
| other records               | +0.01              | +0.48 $\uparrow$   | –0.63 $\downarrow$ | –0.12              | +0.01  | –0.18              | +0.14              | +1.07 $\uparrow$   | –0.35              | +0.21              |

Table S20. Controls for the situation-testing result. Every entry is INTERFAIR minus ERM in percentage points; AUROC is recomputed within the indicated subset.

| Setting              | All   | Unflagged | Random | Low noise | $\Delta$ Acc. |
|----------------------|-------|-----------|--------|-----------|---------------|
| Ohio / MLP           | –0.83 | +0.01     | –0.83  | –0.18     | –0.83         |
| Maryland / MLP       | –0.71 | +0.48     | –0.71  | –0.19     | +0.06         |
| Virginia / MLP       | –1.48 | –0.63     | –1.46  | –0.30     | –1.36         |
| Pennsylvania / MLP   | –0.85 | –0.12     | –0.85  | –0.32     | –0.57         |
| Credit / MLP         | –0.06 | +0.01     | –0.06  | +0.10     | –0.06         |
| ACSIncome / MLP      | –0.79 | –0.18     | –0.78  | –0.54     | +0.24         |
| Employment sex / MLP | –0.95 | +0.14     | –0.95  | –0.36     | –0.14         |
| Employment age / MLP | –1.31 | +1.07     | –1.31  | –0.27     | +2.45         |
| Ohio / FT            | –0.99 | –0.35     | –1.00  | –0.18     | –1.39         |
| ACSIncome / FT       | –0.57 | +0.21     | –0.57  | –0.46     | +0.41         |

Random: mean over 20 size-matched subsets per seed. Low noise: retain labels outside the upper tail of disagreement with own-group neighbors. The accuracy column concerns the unflagged subset. See text for tie handling.

A second control asks whether the flag simply identifies difficult labels. We compute  $N = |y - p_{\text{same}}|$ , disagreement with the record’s own-group neighbors. The upper-tail cutoff for  $N$  uses the same target fraction as situation testing; ties at that cutoff are retained, so the removed counts need not be identical. We then recompute AUROC on the remaining, lower-noise records. This control still has a negative gap in nine settings. By contrast, the unflagged remainder has a negative gap in only four settings, so the situation-testing flag does not simply mark difficult labels.

*Sensitivity and other methods.* Repeating the analysis with 5 or 25 neighbors preserves a larger INTERFAIR–ERM loss on flagged labels in every setting (Table S21). Thresholds .1 and .3 move the boundary of the selected cohort; the export contains each combination of neighborhood size and threshold. Table S22 applies the same test to each published method in all ten settings. LTDD, CoT-Phi, Fair-SMOTE, and reweighing lose more ranking agreement on flagged records in every setting, as does DRAlign in nine of ten; DRAlign’s FT-Transformer models also lose substantially on unflagged records (up to 8.56 points on HMDA). HIFI changes AUROC little, while NeuFair’s loss is evident in both subsets.

Situation testing conditions on observed inputs. Unrecorded legitimate factors can affect the labels, and these data do not identify whether a particular historical decision was justified. The

Table S21. Sensitivity to neighborhood size at situation-testing threshold .2. Each cell gives the AUROC change on flagged / unflagged records, in percentage points (INTERFAIR minus ERM).

| Setting              | $k = 5$       | $k = 10$      | $k = 25$       |
|----------------------|---------------|---------------|----------------|
| Ohio / MLP           | -3.02 / -0.08 | -3.92 / +0.01 | -2.68 / -0.27  |
| Maryland / MLP       | -4.21 / +0.19 | -6.27 / +0.48 | -7.56 / +0.25  |
| Virginia / MLP       | -4.71 / -0.81 | -6.97 / -0.63 | -10.15 / -0.83 |
| Pennsylvania / MLP   | -3.05 / -0.17 | -3.71 / -0.12 | -4.16 / -0.27  |
| Credit / MLP         | -0.41 / -0.02 | -0.75 / +0.01 | -0.71 / -0.03  |
| ACSIIncome / MLP     | -3.06 / -0.16 | -3.88 / -0.18 | -5.28 / -0.18  |
| Employment sex / MLP | -3.05 / +0.00 | -4.14 / +0.14 | -4.95 / +0.06  |
| Employment age / MLP | -6.22 / +0.71 | -7.42 / +1.07 | -8.48 / +1.20  |
| Ohio / FT            | -2.56 / -0.45 | -3.35 / -0.35 | -2.19 / -0.57  |
| ACSIIncome / FT      | -3.51 / +0.20 | -4.59 / +0.21 | -6.07 / +0.14  |

Table S22. Label-cohort comparison for each published method across the ten settings (NeuFair: HMDA Ohio MLP only). Ranges give the smallest and largest ten-seed mean AUROC changes relative to the paper’s ERM, in percentage points.

| Method      | Settings | Flagged range   | Unflagged range | Larger loss |
|-------------|----------|-----------------|-----------------|-------------|
| LTDD        | 10       | [-4.51, -0.48]  | [-0.31, +0.28]  | 10/10       |
| CoT-Phi     | 10       | [-4.74, -0.56]  | [-0.12, +0.20]  | 10/10       |
| DRAlign     | 10       | [-21.24, -0.64] | [-8.56, +1.56]  | 9/10        |
| Fair-SMOTE  | 10       | [-8.36, -0.70]  | [-1.91, +0.73]  | 10/10       |
| Reweighting | 10       | [-3.98, -0.63]  | [-0.05, +0.76]  | 10/10       |
| HIFI        | 10       | [-0.64, +0.34]  | [-0.18, +0.17]  | 7/10        |
| NeuFair     | 1        | -2.74           | -2.02           | 1/1         |

The last column counts settings with a more negative change on flagged records. These are descriptive comparisons against the common ERM, not each pipeline’s own source reference. A range summarizes settings without pooling their records or seeds.

result locates changes in agreement with historical outcomes; it does not provide new ground-truth labels. Accuracy also differs from ranking: Table S20 shows that mortgage accuracy can fall on unflagged records even when AUROC barely changes there.

S7.5 Group error rates

Table S23 expands the mortgage outcome gaps into their component rates. On Ohio MLP, the Black-group false-positive rate rises from .439 to .677: more recorded denials receive approval predictions. Across the five mortgage settings, 88–98% of new label disagreements (ERM correct, INTERFAIR incorrect) are historically denied applications that INTERFAIR approves. HMDA labels record lending decisions rather than repayment, so these rates measure disagreement with past decisions, not default risk.

S7.6 Joint protected attributes

For race and sex jointly, every ordinary edit gives four protected-group responses from eight executions. The equality measure averages the six pairwise response differences. The third-order contrast subtracts the race response contrast under one sex from the same contrast under the other sex. It asks whether the race-dependence of the edit response itself depends on sex. Decision

Table S23. Mortgage error rates against recorded approval and denial labels. Ten-seed means at decision threshold .5. Scott-Knott ESD groups are superscripts; bold marks the best displayed group.

| Method                         | Acc.↑                     | TPR White↑                | TPR Black↑                | FPR White↓                | FPR Black↓                |
|--------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| <i>HMDA Ohio / MLP</i>         |                           |                           |                           |                           |                           |
| ERM                            | <b>0.870</b> <sup>1</sup> | 0.980 <sup>2</sup>        | 0.923 <sup>2</sup>        | <b>0.590</b> <sup>1</sup> | <b>0.439</b> <sup>1</sup> |
| <b>INTERFAIR</b>               | 0.862 <sup>7</sup>        | <b>0.989</b> <sup>1</sup> | <b>0.992</b> <sup>1</sup> | 0.667 <sup>2</sup>        | 0.677 <sup>2</sup>        |
| <i>HMDA Maryland / MLP</i>     |                           |                           |                           |                           |                           |
| ERM                            | <b>0.863</b> <sup>1</sup> | 0.972 <sup>2</sup>        | 0.922 <sup>2</sup>        | <b>0.526</b> <sup>1</sup> | <b>0.439</b> <sup>1</sup> |
| <b>INTERFAIR</b>               | 0.858 <sup>3</sup>        | <b>0.983</b> <sup>1</sup> | <b>0.979</b> <sup>1</sup> | 0.587 <sup>2</sup>        | 0.606 <sup>2</sup>        |
| <i>HMDA Virginia / MLP</i>     |                           |                           |                           |                           |                           |
| ERM                            | <b>0.876</b> <sup>2</sup> | 0.973 <sup>2</sup>        | 0.941 <sup>2</sup>        | <b>0.536</b> <sup>1</sup> | <b>0.480</b> <sup>1</sup> |
| <b>INTERFAIR</b>               | 0.863 <sup>6</sup>        | <b>0.992</b> <sup>1</sup> | <b>0.990</b> <sup>1</sup> | 0.681 <sup>2</sup>        | 0.716 <sup>2</sup>        |
| <i>HMDA Pennsylvania / MLP</i> |                           |                           |                           |                           |                           |
| ERM                            | <b>0.854</b> <sup>2</sup> | 0.963 <sup>2</sup>        | 0.866 <sup>2</sup>        | <b>0.545</b> <sup>1</sup> | <b>0.367</b> <sup>1</sup> |
| <b>INTERFAIR</b>               | 0.847 <sup>6</sup>        | <b>0.981</b> <sup>1</sup> | <b>0.982</b> <sup>1</sup> | 0.632 <sup>2</sup>        | 0.639 <sup>2</sup>        |
| <i>HMDA Ohio / FT</i>          |                           |                           |                           |                           |                           |
| ERM                            | <b>0.873</b> <sup>1</sup> | 0.976 <sup>2</sup>        | 0.933 <sup>2</sup>        | <b>0.564</b> <sup>1</sup> | <b>0.417</b> <sup>1</sup> |
| <b>INTERFAIR</b>               | 0.862 <sup>6</sup>        | <b>0.993</b> <sup>1</sup> | <b>0.994</b> <sup>1</sup> | 0.679 <sup>2</sup>        | 0.702 <sup>2</sup>        |

A false positive is an approval prediction for a recorded denial. These labels do not measure subsequent repayment.

disagreement checks whether all four signed decision transitions agree; direction checks each response. Population penalties average race, sex, and joint-group partitions, and the worst-pair outcome gaps take the largest gap over the six group pairs. Records with unclassified sex remain in task training and race comparisons but are excluded from sex and joint-group terms. Linked income tests must also pass the sex-specific support screens.

Interaction steering penalizes the six pairwise residuals. Pairwise equality across all four responses also bounds their difference-of-differences, so the third-order contrast checks that consequence rather than adding a separate term.

### S7.7 What the component ablation isolates

The ablation in the main paper removes one part of interaction steering at a time while keeping initial weights, task batches, updates, model queries, and selection fixed. Removing equality leaves decision and population terms in place. Removing direction preserves equality on the signed tests. Removing decision terms removes both smooth rate and response penalties. Removing population terms removes the score/effect penalties but retains the separate decision-rate objective. These distinctions explain why decision disagreement can improve without equality, and why the population ablation need not remove every pressure on group outcomes.

### S7.8 Models that pass equality by construction

Two models pass every equality test by construction: one without the protected input, and the additive model  $g(x) + b_a$  of the paper’s problem statement, which uses it only as an offset. Table S24 trains both with task loss only. Neither lowers ERM’s AUROC by more than 0.22 percentage points, and the additive model lowers it by at most 0.14. But neither enforces direction: on HMDA, 18.5% and 20.4% of signed tests still move in the wrong direction, against 0.68% for INTERFAIR. Nor do

Table S24. Two models that pass the equality check by construction, trained with task loss only and validation-AUROC checkpoint selection on ERM’s partitions and initial weights (ten-seed means; Scott–Knott ESD groups compare the three displayed models).  $V$ ,  $D$ , and  $W$  are percentages of tests. Exact-zero ties are bold; undefined groups have no rank.

| Method                      | AUROC $\uparrow$           | Acc. $\uparrow$           | F1 $\uparrow$             | AOD $\downarrow$          | DP $\downarrow$           | EO $\downarrow$           | $\bar{R} \downarrow$ | $V \downarrow$ | $D \downarrow$           | $W \downarrow$            |
|-----------------------------|----------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|----------------------|----------------|--------------------------|---------------------------|
|                             | utility and group outcomes |                           |                           |                           |                           |                           | rule-level behavior  |                |                          |                           |
| <i>HMDA Ohio / MLP</i>      |                            |                           |                           |                           |                           |                           |                      |                |                          |                           |
| ERM                         | <b>0.850</b> <sup>1</sup>  | <b>0.870</b> <sup>1</sup> | <b>0.924</b> <sup>1</sup> | 0.104 <sup>3</sup>        | 0.149 <sup>2</sup>        | 0.151 <sup>3</sup>        | 0.200                | 80.73          | 5.34 <sup>3</sup>        | 26.13 <sup>3</sup>        |
| Protected input removed     | <b>0.850</b> <sup>1</sup>  | 0.870 <sup>2</sup>        | <b>0.924</b> <sup>1</sup> | <b>0.051</b> <sup>1</sup> | <b>0.102</b> <sup>1</sup> | <b>0.079</b> <sup>1</sup> | <b>0.000</b>         | <b>0.00</b>    | <b>0.00</b> <sup>1</sup> | <b>18.48</b> <sup>1</sup> |
| Additive model $g(x) + b_a$ | <b>0.850</b> <sup>1</sup>  | 0.870 <sup>2</sup>        | 0.923 <sup>2</sup>        | 0.098 <sup>2</sup>        | 0.146 <sup>2</sup>        | 0.142 <sup>2</sup>        | <b>0.000</b>         | <b>0.00</b>    | 3.21 <sup>2</sup>        | 20.43 <sup>2</sup>        |
| <i>Credit / MLP</i>         |                            |                           |                           |                           |                           |                           |                      |                |                          |                           |
| ERM                         | 0.785 <sup>2</sup>         | 0.820 <sup>3</sup>        | 0.891 <sup>2</sup>        | 0.025 <sup>3</sup>        | 0.032 <sup>3</sup>        | 0.033 <sup>3</sup>        | 0.040                | 24.43          | 1.90 <sup>3</sup>        | 6.62 <sup>2</sup>         |
| Protected input removed     | 0.785 <sup>2</sup>         | <b>0.821</b> <sup>1</sup> | <b>0.892</b> <sup>1</sup> | <b>0.010</b> <sup>1</sup> | <b>0.021</b> <sup>1</sup> | <b>0.013</b> <sup>1</sup> | <b>0.000</b>         | <b>0.00</b>    | <b>0.00</b> <sup>1</sup> | <b>4.29</b> <sup>1</sup>  |
| Additive model $g(x) + b_a$ | <b>0.786</b> <sup>1</sup>  | 0.820 <sup>2</sup>        | <b>0.892</b> <sup>1</sup> | 0.016 <sup>2</sup>        | 0.026 <sup>2</sup>        | 0.020 <sup>2</sup>        | <b>0.000</b>         | <b>0.00</b>    | 1.14 <sup>2</sup>        | 4.52 <sup>1</sup>         |
| <i>ACSIncome / MLP</i>      |                            |                           |                           |                           |                           |                           |                      |                |                          |                           |
| ERM                         | <b>0.905</b> <sup>1</sup>  | 0.819 <sup>2</sup>        | 0.815 <sup>2</sup>        | 0.070 <sup>3</sup>        | 0.152 <sup>3</sup>        | 0.081 <sup>3</sup>        | 0.107                | 63.90          | 13.82 <sup>3</sup>       | 0.13 <sup>2</sup>         |
| Protected input removed     | 0.902 <sup>3</sup>         | 0.817 <sup>3</sup>        | 0.813 <sup>3</sup>        | <b>0.027</b> <sup>1</sup> | <b>0.109</b> <sup>1</sup> | <b>0.039</b> <sup>1</sup> | <b>0.000</b>         | <b>0.00</b>    | <b>0.00</b> <sup>1</sup> | <b>0.11</b> <sup>1</sup>  |
| Additive model $g(x) + b_a$ | 0.903 <sup>2</sup>         | <b>0.821</b> <sup>1</sup> | <b>0.818</b> <sup>1</sup> | 0.041 <sup>2</sup>        | 0.124 <sup>2</sup>        | 0.053 <sup>2</sup>        | <b>0.000</b>         | <b>0.00</b>    | 4.66 <sup>2</sup>        | <b>0.11</b> <sup>1</sup>  |

they narrow group gaps as much: on all three tasks, INTERFAIR has lower AOD and DP than both, at AUROC costs of 0.06–0.83 percentage points, and all six DP differences survive Holm correction. The additive model’s offset also still produces decision-response disagreement (3.21% on HMDA). Removal does not stop the rule from depending on proxies: when a test flips the tract’s minority share between the White and Black applicants’ medians (8% and 39%) instead of race, the effects of the HMDA features depend on it as much in the model without race as in ERM ( $\bar{R}$  .235 vs. .242), whereas INTERFAIR lowers this dependence to .074. The paper’s discussion considers when removal is the simpler repair.

Table S24 compares these delivered models with the primary ERM, whose validation schedule differs. A separate matched task-only experiment, with its own ERM and an identical schedule (40 epochs, AdamW at .0007, weight decay .0001, batch 512, clipping 5, and per-epoch validation-AUROC selection), isolates the fit cost of the additive restriction. ERM’s AUROC advantage is .043, .007, and .080 percentage points on HMDA, Credit, and ACSIncome; only the last is significant after correction ( $p = .00112$ ), and mean log-loss differences are at most .000162 in magnitude.

## S8 RQ4 (Regression and robustness): Edits, populations, updates, and sensitivity

This section supports the paper’s RQ4: whether the saved tests catch regressions after updates and whether the repaired rule holds on new edits and new populations. It ends with the sensitivity of the repair to its training edits, its training pool, and the audit size.

### S8.1 Larger and new edits

The main paper reports linked income increments of \$20k, \$30k, and \$50k. For the additional HMDA states, supported counts at \$10k/\$20k/\$30k/\$50k are 737/726/717/661 in Maryland, 716/704/684/583 in Virginia, and 656/647/621/496 in Pennsylvania. Each method uses identical supported profiles at a given increment, with valid reported DTI codes.

The additional census audit changes education from high-school completion to bachelor’s degree (codes 16 to 21) or bachelor’s to master’s degree (21 to 22), and increases weekly hours by five or ten.

Table S25. Equality and decision responses on additional edit values. Means with sample standard deviations over ten seeds;  $V$  and  $D$  are percentages. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Method                  | $\bar{R} \downarrow$                       | $V \downarrow$                           | $D \downarrow$                           |
|-------------------------|--------------------------------------------|------------------------------------------|------------------------------------------|
| <i>Credit / MLP</i>     |                                            |                                          |                                          |
| ERM                     | 0.036 <sup>2</sup> <sub>±0.025</sub>       | 21.93 <sup>2</sup> <sub>±17.17</sub>     | 1.83 <sup>2</sup> <sub>±0.83</sub>       |
| <b>INTERFAIR</b>        | <b>0.009<sup>1</sup></b> <sub>±0.001</sub> | <b>0.20<sup>1</sup></b> <sub>±0.12</sub> | <b>0.51<sup>1</sup></b> <sub>±0.32</sub> |
| <i>ACSIIncome / MLP</i> |                                            |                                          |                                          |
| ERM                     | 0.117 <sup>2</sup> <sub>±0.020</sub>       | 66.30 <sup>2</sup> <sub>±6.30</sub>      | 14.58 <sup>2</sup> <sub>±1.26</sub>      |
| <b>INTERFAIR</b>        | <b>0.012<sup>1</sup></b> <sub>±0.003</sub> | <b>1.40<sup>1</sup></b> <sub>±1.29</sub> | <b>0.44<sup>1</sup></b> <sub>±0.11</sub> |
| <i>ACSIIncome / FT</i>  |                                            |                                          |                                          |
| ERM                     | 0.223 <sup>2</sup> <sub>±0.045</sub>       | 77.42 <sup>2</sup> <sub>±3.18</sub>      | 13.52 <sup>2</sup> <sub>±1.55</sub>      |
| <b>INTERFAIR</b>        | <b>0.006<sup>1</sup></b> <sub>±0.002</sub> | <b>0.02<sup>1</sup></b> <sub>±0.02</sub> | <b>0.29<sup>1</sup></b> <sub>±0.06</sub> |

Direction is defined only for the added ACSIIncome education and hours edits. Their wrong-direction rates are reported separately in the text.

Hour endpoints must stay inside both schema bounds and the training 1st/99th-percentile range, 4–75 hours. Categorical contrasts include less frequent work-class, household-status, and occupation levels. Household edits change marital status and relationship jointly and retain spouse-consistency checks. These tests exercise values beyond the training endpoint pair.

Credit’s additional edits compare its most frequent and third most frequent education, marriage, and payment-status categories. They receive equality checks but no direction. Table S25 therefore compares equality and decision behavior on both tasks, while direction is evaluated separately for the census education and hours edits. On those signed edits, ACSIIncome MLP’s  $W$  rises from 3.34% to 4.30%, whereas FT’s falls from 3.14% to 2.01%. Generalization of equality and decision behavior thus does not ensure generalization of every supplied direction.

## S8.2 Geographic and temporal transfer

Ohio 2019 checkpoints are applied unchanged to 12,000 Michigan 2019 and 12,000 Ohio 2020 applications. Model parameters, source encoders, endpoints, support rules, decision thresholds, and checkpoint choices remain frozen. No destination labels are used for fitting or selection. The main paper gives the MLP comparison. Tables S26 and S27 give the deferred FT-Transformer results with the same utility, outcome, and behavioral measures. In both destinations, DRAlign has the lowest interaction residual and group-outcome gaps, while INTERFAIR has the lowest decision-response disagreement and wrong-direction rate. INTERFAIR retains higher AUROC: .826 and .827, compared with .729 and .731 for DRAlign.

## S8.3 Regression testing during maintenance

Every HMDA Ohio INTERFAIR model receives ten task-only epochs at learning rate .0001, retaining the final updated checkpoint. Two follow-ups start from that checkpoint: ten more task-only epochs or ten epochs of interaction steering, both at .0001. They share task draws, contexts, update and query budgets, and validation opportunities. Replaying the same held-out tests detects whether the update still passes the checks and whether the follow-up repairs a regression.

Table S26. FT-Transformer models trained in Ohio 2019 and evaluated without refitting. Ten-seed means; Scott-Knott ESD ranks are superscripts; bold marks the best displayed group.

| Method                                | AUROC↑                    | Acc.↑                     | F1↑                       | AOD↓                      | DP↓                       | EO↓                       |
|---------------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| <i>Michigan 2019 / FT-Transformer</i> |                           |                           |                           |                           |                           |                           |
| ERM                                   | <b>0.844</b> <sup>1</sup> | <b>0.874</b> <sup>1</sup> | <b>0.926</b> <sup>1</sup> | 0.054 <sup>5</sup>        | 0.119 <sup>7</sup>        | 0.063 <sup>6</sup>        |
| LTDD                                  | 0.838 <sup>5</sup>        | 0.872 <sup>4</sup>        | 0.925 <sup>3</sup>        | 0.032 <sup>4</sup>        | 0.063 <sup>4</sup>        | 0.048 <sup>4</sup>        |
| CoT-Phi                               | 0.841 <sup>3</sup>        | 0.873 <sup>2</sup>        | 0.926 <sup>2</sup>        | 0.022 <sup>2</sup>        | 0.073 <sup>5</sup>        | 0.031 <sup>2</sup>        |
| DRAlign                               | 0.729 <sup>9</sup>        | 0.755 <sup>8</sup>        | 0.809 <sup>8</sup>        | <b>0.006</b> <sup>1</sup> | <b>0.011</b> <sup>1</sup> | <b>0.011</b> <sup>1</sup> |
| Fair-SMOTE                            | 0.815 <sup>8</sup>        | 0.810 <sup>7</sup>        | 0.880 <sup>7</sup>        | 0.070 <sup>6</sup>        | 0.077 <sup>5</sup>        | 0.104 <sup>7</sup>        |
| Reweighing                            | 0.840 <sup>4</sup>        | 0.873 <sup>3</sup>        | 0.926 <sup>2</sup>        | 0.030 <sup>3</sup>        | 0.049 <sup>3</sup>        | 0.057 <sup>5</sup>        |
| HIFI                                  | 0.843 <sup>2</sup>        | <b>0.874</b> <sup>1</sup> | <b>0.926</b> <sup>1</sup> | 0.054 <sup>5</sup>        | 0.118 <sup>7</sup>        | 0.066 <sup>6</sup>        |
| <b>INTERFAIR</b>                      | 0.826 <sup>6</sup>        | 0.866 <sup>5</sup>        | 0.923 <sup>4</sup>        | 0.022 <sup>2</sup>        | 0.039 <sup>2</sup>        | 0.040 <sup>3</sup>        |
| <i>Ohio 2020 / FT-Transformer</i>     |                           |                           |                           |                           |                           |                           |
| ERM                                   | <b>0.840</b> <sup>1</sup> | <b>0.897</b> <sup>1</sup> | <b>0.942</b> <sup>1</sup> | 0.087 <sup>7</sup>        | 0.097 <sup>6</sup>        | 0.148 <sup>7</sup>        |
| LTDD                                  | 0.837 <sup>3</sup>        | 0.896 <sup>3</sup>        | 0.941 <sup>3</sup>        | 0.035 <sup>4</sup>        | 0.060 <sup>4</sup>        | 0.063 <sup>3</sup>        |
| CoT-Phi                               | 0.836 <sup>4</sup>        | 0.896 <sup>3</sup>        | 0.941 <sup>3</sup>        | 0.050 <sup>5</sup>        | 0.073 <sup>5</sup>        | 0.085 <sup>4</sup>        |
| DRAlign                               | 0.731 <sup>9</sup>        | 0.782 <sup>8</sup>        | 0.827 <sup>8</sup>        | <b>0.008</b> <sup>1</sup> | <b>0.009</b> <sup>1</sup> | <b>0.009</b> <sup>1</sup> |
| Fair-SMOTE                            | 0.815 <sup>8</sup>        | 0.857 <sup>7</sup>        | 0.915 <sup>7</sup>        | 0.048 <sup>5</sup>        | 0.094 <sup>6</sup>        | 0.059 <sup>3</sup>        |
| Reweighing                            | 0.837 <sup>3</sup>        | 0.896 <sup>2</sup>        | 0.942 <sup>2</sup>        | 0.032 <sup>3</sup>        | 0.056 <sup>3</sup>        | 0.060 <sup>3</sup>        |
| HIFI                                  | 0.840 <sup>2</sup>        | 0.896 <sup>2</sup>        | 0.941 <sup>2</sup>        | 0.086 <sup>7</sup>        | 0.097 <sup>6</sup>        | 0.144 <sup>6</sup>        |
| <b>INTERFAIR</b>                      | 0.827 <sup>6</sup>        | 0.889 <sup>5</sup>        | 0.938 <sup>5</sup>        | 0.012 <sup>2</sup>        | 0.035 <sup>2</sup>        | 0.022 <sup>2</sup>        |

Table S27. FT-Transformer models trained in Ohio 2019 and evaluated without refitting. Ten-seed means; Scott-Knott ESD ranks are superscripts; bold marks the best displayed group.  $V$ ,  $D$ , and  $W$  are percentages.

| Method                                | $\bar{R}$ ↓               | $V$ ↓                    | $D$ ↓                    | $W$ ↓                    |
|---------------------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| <i>Michigan 2019 / FT-Transformer</i> |                           |                          |                          |                          |
| ERM                                   | 0.203 <sup>6</sup>        | 73.80 <sup>7</sup>       | 4.12 <sup>5</sup>        | 25.35 <sup>7</sup>       |
| LTDD                                  | 0.303 <sup>8</sup>        | 84.51 <sup>9</sup>       | 4.23 <sup>6</sup>        | 25.40 <sup>7</sup>       |
| CoT-Phi                               | 0.291 <sup>7</sup>        | 82.06 <sup>8</sup>       | 3.55 <sup>4</sup>        | 19.50 <sup>5</sup>       |
| DRAlign                               | <b>0.002</b> <sup>1</sup> | <b>0.01</b> <sup>1</sup> | 0.31 <sup>2</sup>        | 5.82 <sup>2</sup>        |
| Fair-SMOTE                            | 0.407 <sup>9</sup>        | 86.86 <sup>10</sup>      | 11.22 <sup>8</sup>       | 40.99 <sup>8</sup>       |
| Reweighing                            | 0.090 <sup>3</sup>        | 44.46 <sup>3</sup>       | 2.23 <sup>3</sup>        | 17.26 <sup>4</sup>       |
| HIFI                                  | 0.189 <sup>5</sup>        | 70.82 <sup>6</sup>       | 4.34 <sup>6</sup>        | 22.44 <sup>6</sup>       |
| <b>INTERFAIR</b>                      | 0.005 <sup>2</sup>        | 0.02 <sup>2</sup>        | <b>0.13</b> <sup>1</sup> | <b>0.17</b> <sup>1</sup> |
| <i>Ohio 2020 / FT-Transformer</i>     |                           |                          |                          |                          |
| ERM                                   | 0.223 <sup>6</sup>        | 75.42 <sup>7</sup>       | 3.73 <sup>7</sup>        | 22.96 <sup>7</sup>       |
| LTDD                                  | 0.306 <sup>7</sup>        | 84.68 <sup>9</sup>       | 3.57 <sup>6</sup>        | 22.62 <sup>7</sup>       |
| CoT-Phi                               | 0.299 <sup>7</sup>        | 82.49 <sup>8</sup>       | 3.20 <sup>5</sup>        | 17.31 <sup>5</sup>       |
| DRAlign                               | <b>0.002</b> <sup>1</sup> | <b>0.01</b> <sup>1</sup> | 0.30 <sup>2</sup>        | 5.97 <sup>2</sup>        |
| Fair-SMOTE                            | 0.389 <sup>8</sup>        | 86.11 <sup>10</sup>      | 9.13 <sup>9</sup>        | 36.01 <sup>8</sup>       |
| Reweighing                            | 0.090 <sup>3</sup>        | 44.66 <sup>3</sup>       | 1.81 <sup>3</sup>        | 15.29 <sup>4</sup>       |
| HIFI                                  | 0.208 <sup>5</sup>        | 72.73 <sup>6</sup>       | 4.02 <sup>8</sup>        | 20.58 <sup>6</sup>       |
| <b>INTERFAIR</b>                      | 0.005 <sup>2</sup>        | 0.01 <sup>2</sup>        | <b>0.10</b> <sup>1</sup> | <b>0.10</b> <sup>1</sup> |

Table S28. Replaying the HMDA Ohio tests through a model update and two matched follow-up fits. Ten-seed means; rate columns are percentages. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Training                     | AUROC↑                    | AOD↓                      | $\bar{R}$ ↓               | V ↓                      | D ↓                      | W ↓                      |
|------------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| INTERFAIR before update      | 0.842 <sup>4</sup>        | 0.008 <sup>2</sup>        | <b>0.008</b> <sup>1</sup> | <b>0.10</b> <sup>1</sup> | 0.13 <sup>2</sup>        | <b>0.68</b> <sup>1</sup> |
| After task-only update       | 0.849 <sup>2</sup>        | 0.097 <sup>3</sup>        | 0.058 <sup>3</sup>        | 38.49 <sup>3</sup>       | 4.12 <sup>3</sup>        | 9.25 <sup>3</sup>        |
| Further task-only training   | <b>0.850</b> <sup>1</sup> | 0.111 <sup>4</sup>        | 0.072 <sup>4</sup>        | 47.36 <sup>4</sup>       | 4.99 <sup>4</sup>        | 12.19 <sup>4</sup>       |
| Further interaction steering | 0.847 <sup>3</sup>        | <b>0.006</b> <sup>1</sup> | 0.012 <sup>2</sup>        | 0.88 <sup>2</sup>        | <b>0.10</b> <sup>1</sup> | 1.72 <sup>2</sup>        |

Table S29. Choosing training edits on HMDA Ohio. All repairs are evaluated on the same supported p20/p80 audit specification; ten-seed means. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Training             | AUROC↑                    | AOD↓                      | $\bar{R}$ ↓               | V ↓                      | D ↓                      | W ↓                      |
|----------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| ERM                  | <b>0.850</b> <sup>1</sup> | 0.104 <sup>5</sup>        | 0.200 <sup>5</sup>        | 80.73 <sup>5</sup>       | 5.34 <sup>4</sup>        | 26.13 <sup>4</sup>       |
| p20/p80, supported   | 0.845 <sup>4</sup>        | 0.010 <sup>2</sup>        | <b>0.006</b> <sup>1</sup> | <b>0.06</b> <sup>1</sup> | <b>0.10</b> <sup>1</sup> | <b>0.83</b> <sup>1</sup> |
| p20/p80, schema only | 0.845 <sup>4</sup>        | <b>0.008</b> <sup>1</sup> | <b>0.006</b> <sup>1</sup> | <b>0.05</b> <sup>1</sup> | <b>0.11</b> <sup>1</sup> | <b>0.78</b> <sup>1</sup> |
| p10/p90, supported   | 0.838 <sup>5</sup>        | 0.011 <sup>3</sup>        | 0.009 <sup>2</sup>        | 0.12 <sup>2</sup>        | 0.14 <sup>2</sup>        | <b>0.87</b> <sup>1</sup> |
| p35/p65, supported   | 0.848 <sup>2</sup>        | 0.009 <sup>2</sup>        | 0.009 <sup>3</sup>        | 0.26 <sup>3</sup>        | 0.15 <sup>2</sup>        | 2.57 <sup>2</sup>        |
| Local, supported     | 0.847 <sup>3</sup>        | 0.013 <sup>4</sup>        | 0.014 <sup>4</sup>        | 1.73 <sup>4</sup>        | 0.21 <sup>3</sup>        | 9.28 <sup>3</sup>        |

Table S28 gives the complete sequence behind the paper’s maintenance figure, including utility and AOD. Renewed interaction steering brings AOD and  $D$  below their pre-update values and  $\bar{R}$ ,  $V$ , and  $W$  most of the way back, at AUROC .847 against .850 for further task-only training.

#### S8.4 Training-edit choice

To assess the default endpoints, the HMDA experiment runs interaction steering with supported p20/p80, schema-valid p20/p80 without the proximity screen, supported p10/p90, supported p35/p65, or local observed-value edits. All arms use the matched update and query budget of Section S6, and all are evaluated on the same primary supported banks. Thus the comparison tests training-edit choice without changing the evaluation population.

A local numerical edit starts at the record’s observed value  $v$  and takes the smallest observed training value at least  $v + h$ , where  $h = \max\{(q_{.8} - q_{.2})/20, 10^{-6}\}$  in encoded units. Missing starting values and values with no larger endpoint are ineligible. Categorical endpoints are unchanged. All policies retain linked +\$10k income steering; larger income increments remain held out.

Table S29 supports broad training contrasts over these small local edits. Local training leaves 9.3% wrong-direction tests, compared with about .8% for the default. The default provides a broad, reproducible contrast, and the held-out edit tests assess behavior outside its endpoints.

#### S8.5 Training-pool and audit-size sensitivity

The training-pool study varies how many distinct contexts can supply the penalties: 512, 2,048, or all eligible contexts. The limit applies to natural population records and edited contexts. Every arm keeps the number of task updates and sampled contexts fixed. Table S30 shows that finite pools can support substantial repair, while population gaps and rare direction failures still depend on context diversity.

Table S30. Distinct training contexts at fixed compute. Task updates and numbers of sampled contexts are identical; only the pools from which they are sampled change. Ten-seed means; rates are percentages. Scott-Knott ESD groups are superscripts; bold marks the best displayed group.

| Pool                    | AUROC $\uparrow$          | AOD $\downarrow$          | $\bar{R}$ $\downarrow$    | V $\downarrow$           | D $\downarrow$           | W $\downarrow$           |
|-------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| <i>HMDA Ohio / MLP</i>  |                           |                           |                           |                          |                          |                          |
| 512 contexts            | <b>0.846</b> <sup>2</sup> | 0.023 <sup>2</sup>        | <b>0.006</b> <sup>1</sup> | <b>0.06</b> <sup>1</sup> | <b>0.11</b> <sup>1</sup> | 1.43 <sup>3</sup>        |
| 2,048 contexts          | 0.844 <sup>4</sup>        | <b>0.011</b> <sup>1</sup> | 0.007 <sup>2</sup>        | 0.10 <sup>2</sup>        | 0.14 <sup>2</sup>        | 0.88 <sup>2</sup>        |
| All eligible contexts   | 0.845 <sup>3</sup>        | <b>0.011</b> <sup>1</sup> | <b>0.006</b> <sup>1</sup> | <b>0.05</b> <sup>1</sup> | <b>0.11</b> <sup>1</sup> | <b>0.71</b> <sup>1</sup> |
| <i>ACSIIncome / MLP</i> |                           |                           |                           |                          |                          |                          |
| 512 contexts            | <b>0.900</b> <sup>2</sup> | 0.025 <sup>3</sup>        | 0.012 <sup>3</sup>        | 0.44 <sup>2</sup>        | 0.70 <sup>2</sup>        | 0.12 <sup>3</sup>        |
| 2,048 contexts          | <b>0.899</b> <sup>2</sup> | 0.017 <sup>2</sup>        | 0.011 <sup>2</sup>        | 0.49 <sup>2</sup>        | 0.90 <sup>3</sup>        | 0.10 <sup>2</sup>        |
| All eligible contexts   | 0.898 <sup>3</sup>        | <b>0.014</b> <sup>1</sup> | <b>0.009</b> <sup>1</sup> | <b>0.12</b> <sup>1</sup> | <b>0.60</b> <sup>1</sup> | <b>0.06</b> <sup>1</sup> |

Audit size is a separate issue. On the 70 paired ERM/INTERFAIR checkpoints from the seven task-architecture settings, 30 seeded permutations produce nested subsets of 64, 128, 256, 512, and 1,024 supported contexts per relation, capped by the available bank. Both models use identical subsets. The benchmark is the corresponding difference on the full finite bank, not an assumed population truth.

At 256 contexts per relation, all 2,100 subsamples preserve the full-bank direction of the residual and decision-disagreement differences. Mean absolute error in the decision-disagreement difference ranges from .15 to .73 percentage points across settings. Rare direction failures need larger banks: for ACSIIncome MLP, agreement with the sign of the full-bank per-seed  $W$  difference is 44.7% at 256 contexts and 78.3% at 1,024. The paper uses full-bank results. This experiment concerns measurement precision for fixed models; repeated optimizer seeds address a different source of variation.

## S9 Additional comparison methods and model families

These comparisons have a different prediction interface or a smaller evaluation scope than the paper’s main comparison.

### S9.1 Decision-only and randomized output rules

MirrorFair [15] combines a source network with a mirror network trained on protected-mutated records. The mirror fit uses 40 epochs, AdamW at .001, decay .0001, batch 512, and mirrored-validation AUROC selection; the native combination scenario is chosen on natural validation records. Its final interface returns a hard decision, so we report decision disagreement and conventional decision metrics (Table S31). Assigning logits to those decisions would create an arbitrary score scale and is unnecessary for testing their behavior.

Equalized-odds post-processing [4] fits a group-dependent threshold rule on source validation labels. We use the threshold optimizer with accuracy objective, grid size 1,000, and flip=False. For a randomized threshold mixture, each of the four executions returns its exact probability of acceptance. Independent random draws at the four inputs would otherwise add disagreement caused by sampling the postprocessor rather than by changing the input.

The residual computed on acceptance probabilities instead of log-odds is denoted  $R_{\text{accept}}$ . This is a property of the output decision rule, distinct from the underlying logit residual. Accuracy and F1 use expected confusion counts. Table S32 shows that improving equalized odds does not force equal edit responses of the resulting acceptance rule.

Table S31. MirrorFair’s native decision interface on the primary audit banks. Ten-seed means;  $D$  is a percentage. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Method                 | Acc.↑                    | F1↑                      | AOD↓                     | DP↓                      | EO↓                      | $D$ ↓                   |
|------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------|
| <i>HMDA Ohio / MLP</i> |                          |                          |                          |                          |                          |                         |
| ERM                    | <b>0.870<sup>1</sup></b> | <b>0.924<sup>1</sup></b> | 0.104 <sup>8</sup>       | 0.149 <sup>7</sup>       | 0.151 <sup>7</sup>       | 5.34 <sup>5</sup>       |
| MirrorFair             | 0.870 <sup>2</sup>       | 0.923 <sup>2</sup>       | <b>0.017<sup>3</sup></b> | <b>0.072<sup>4</sup></b> | <b>0.026<sup>2</sup></b> | <b>3.16<sup>3</sup></b> |
| <i>Credit / MLP</i>    |                          |                          |                          |                          |                          |                         |
| ERM                    | 0.820 <sup>2</sup>       | 0.891 <sup>3</sup>       | 0.025 <sup>6</sup>       | 0.032 <sup>7</sup>       | 0.033 <sup>7</sup>       | <b>1.90<sup>5</sup></b> |
| MirrorFair             | <b>0.820<sup>1</sup></b> | <b>0.892<sup>1</sup></b> | <b>0.016<sup>5</sup></b> | <b>0.006<sup>1</sup></b> | <b>0.029<sup>6</sup></b> | <b>1.90<sup>5</sup></b> |
| <i>ACSIncome / MLP</i> |                          |                          |                          |                          |                          |                         |
| ERM                    | <b>0.819<sup>2</sup></b> | <b>0.815<sup>2</sup></b> | 0.070 <sup>8</sup>       | 0.152 <sup>11</sup>      | 0.081 <sup>8</sup>       | 13.82 <sup>8</sup>      |
| MirrorFair             | 0.814 <sup>5</sup>       | 0.812 <sup>3</sup>       | <b>0.036<sup>3</sup></b> | <b>0.047<sup>2</sup></b> | <b>0.048<sup>4</sup></b> | <b>7.50<sup>5</sup></b> |

MirrorFair returns hard decisions. A logit equality residual and a probability-response magnitude are therefore not defined for its native interface.

Table S32. Fitted equalized-odds post-processing. Ten-seed means;  $\bar{R}_{\text{accept}}$  is the mean absolute residual of the acceptance probability. Scott–Knott ESD groups are superscripts; bold marks the best displayed group.

| Task      | Rule          | Acc.↑                   | F1↑                     | AOD↓                    | DP↓                     | $\bar{R}_{\text{accept}}$ ↓ |
|-----------|---------------|-------------------------|-------------------------|-------------------------|-------------------------|-----------------------------|
| HMDA Ohio | ERM           | <b>.870<sup>1</sup></b> | <b>.923<sup>1</sup></b> | .104 <sup>2</sup>       | .148 <sup>2</sup>       | .060 <sup>2</sup>           |
|           | EO processing | .867 <sup>2</sup>       | .923 <sup>2</sup>       | <b>.024<sup>1</sup></b> | <b>.041<sup>1</sup></b> | <b>.026<sup>1</sup></b>     |
| Credit    | ERM           | <b>.821<sup>1</sup></b> | <b>.892<sup>1</sup></b> | .027 <sup>2</sup>       | .033 <sup>2</sup>       | <b>.018<sup>1</sup></b>     |
|           | EO processing | .807 <sup>2</sup>       | .883 <sup>2</sup>       | <b>.021<sup>1</sup></b> | <b>.010<sup>1</sup></b> | .023 <sup>2</sup>           |
| ACSIncome | ERM           | <b>.819<sup>1</sup></b> | <b>.815<sup>1</sup></b> | .067 <sup>2</sup>       | .149 <sup>2</sup>       | .135 <sup>2</sup>           |
|           | EO processing | .811 <sup>2</sup>       | .805 <sup>2</sup>       | <b>.017<sup>1</sup></b> | <b>.065<sup>1</sup></b> | <b>.058<sup>1</sup></b>     |

Table S33. DICE neuron interventions on HMDA Ohio MLP, two-seed descriptive means; bold marks the best observed mean, with no significance grouping.  $D$  is a percentage.

| Method              | AUROC↑      | AOD↓        | DP↓  | $\bar{R}$ ↓ | $D$ ↓       |
|---------------------|-------------|-------------|------|-------------|-------------|
| Source ERM          | .851        | .105        | .145 | .190        | 6.23        |
| DICE, force minimum | <b>.851</b> | .105        | .145 | <b>.187</b> | 6.21        |
| DICE, force mean    | .851        | <b>.103</b> | .142 | .187        | <b>6.03</b> |

## S9.2 Neuron intervention and additional architectures

The DICE comparison [10] uses two HMDA seeds, its actual source ERM, and both released neuron-force interventions. The adapter supplies validation candidates to the neuron-selection procedure; it does not reproduce the full native search. The source validation partition supplies the candidates. Table S33 reports these descriptive outcomes separately from the ten-seed method comparisons.

The audit needs only model queries. Table S34 demonstrates it on histogram gradient boosting and a residual MLP. The tree audits use ordinary endpoint contrasts; the Credit tree study covers 13 monetary fields rather than the main specification’s 21 fields. The residual MLP uses a \$10k income increment with DTI held fixed. These different test sets demonstrate interface portability and must not be read as an architecture ranking. Repair itself is evaluated on the MLP and FT-Transformer settings of the paper.

Table S34. Additional audit interfaces. Means over the stated number of seeds; rates are percentages.

| Task      | Model              | Seeds | AUROC | $\bar{R}$ | $V$   | $D$  |
|-----------|--------------------|-------|-------|-----------|-------|------|
| HMDA Ohio | Histogram boosting | 10    | .859  | .162      | 63.60 | 3.77 |
| Credit    | Histogram boosting | 10    | .787  | .050      | 30.61 | 1.36 |
| HMDA Ohio | Residual MLP       | 2     | .848  | .030      | 18.82 | .56  |

S9.3 NeuFair on HMDA Ohio

NeuFair [2] searches neuron masks for one hour per seed (Table S6), so we ran it on HMDA Ohio only and report it here rather than in the paper’s main comparison, which covers the six methods available in every setting. Against its source model, the same unmitigated MLP that the LTDD, CoT-Phi, and DRAlign reference comparisons use, NeuFair narrows AOD by 69% (.104 to .032) and lowers the rule violation by 47% (.183 to .097), a decrease that survives Holm correction. Against the common ERM it leaves more signed tests moving in the wrong direction (35.80% against 26.13%), and its Ohio models keep the same pattern in Michigan 2019 and Ohio 2020 (Table S35).

S10 Replication material

The replication package contains data-preparation scripts, prepared inputs and edit specifications, training configurations, model adapters, and per-seed measurements. It also includes mean/standard-deviation summaries, paired-test results, Scott–Knott inputs and diagnostics, selected-epoch records, and scripts for regenerating the primary, transfer, and attribute-flip tables and the numerical figures from the supplied measurements.

The result index maps supporting studies to their measurement and analysis files, and the manifest records file hashes. The label-cohort exports report the per-seed subset comparisons and neighborhood/threshold sensitivity results. The package provides training code; fitted model checkpoints and per-record model predictions are not included.

Table S35. NeuFair on HMDA Ohio and its transfer without refitting, beside ERM and INTERFAIR from the paper’s tables (ten-seed means; Scott–Knott ESD groups from the paper’s complete comparison families).

| Method                                                  | AUROC↑                    | Acc.↑                     | F1↑                       | AOD↓                      | DP↓                       | EO↓                       | $\bar{R}$ ↓               | $V$ ↓                    | $D$ ↓                    | $W$ ↓                    |
|---------------------------------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|
| utility and group outcomes                              |                           |                           |                           |                           |                           |                           | rule-level behavior       |                          |                          |                          |
| <i>HMDA Ohio 2019 / MLP</i>                             |                           |                           |                           |                           |                           |                           |                           |                          |                          |                          |
| ERM                                                     | <b>0.850</b> <sup>1</sup> | <b>0.870</b> <sup>1</sup> | <b>0.924</b> <sup>1</sup> | 0.104 <sup>8</sup>        | 0.149 <sup>7</sup>        | 0.151 <sup>7</sup>        | 0.200 <sup>4</sup>        | 80.73 <sup>3</sup>       | 5.34 <sup>5</sup>        | 26.13 <sup>5</sup>       |
| NeuFair                                                 | 0.829 <sup>6</sup>        | 0.851 <sup>9</sup>        | 0.915 <sup>7</sup>        | 0.032 <sup>5</sup>        | <b>0.022</b> <sup>1</sup> | 0.060 <sup>4</sup>        | 0.097 <sup>2</sup>        | 62.04 <sup>2</sup>       | 2.08 <sup>2</sup>        | 35.80 <sup>8</sup>       |
| <b>INTERFAIR</b>                                        | 0.842 <sup>4</sup>        | 0.862 <sup>7</sup>        | 0.920 <sup>5</sup>        | <b>0.008</b> <sup>1</sup> | 0.044 <sup>2</sup>        | <b>0.014</b> <sup>1</sup> | <b>0.008</b> <sup>1</sup> | <b>0.10</b> <sup>1</sup> | <b>0.13</b> <sup>1</sup> | <b>0.68</b> <sup>1</sup> |
| <i>Michigan 2019 / MLP (Ohio models, no refitting)</i>  |                           |                           |                           |                           |                           |                           |                           |                          |                          |                          |
| ERM                                                     | <b>0.831</b> <sup>1</sup> | <b>0.870</b> <sup>1</sup> | <b>0.924</b> <sup>1</sup> | 0.063 <sup>7</sup>        | 0.126 <sup>8</sup>        | 0.072 <sup>6</sup>        | 0.199 <sup>4</sup>        | 80.56 <sup>3</sup>       | 5.15 <sup>5</sup>        | 25.32 <sup>5</sup>       |
| NeuFair                                                 | 0.807 <sup>7</sup>        | 0.855 <sup>10</sup>       | 0.918 <sup>6</sup>        | 0.038 <sup>4</sup>        | <b>0.018</b> <sup>1</sup> | 0.071 <sup>6</sup>        | 0.096 <sup>2</sup>        | 61.77 <sup>2</sup>       | 1.90 <sup>2</sup>        | 34.69 <sup>7</sup>       |
| <b>INTERFAIR</b>                                        | 0.819 <sup>6</sup>        | 0.865 <sup>7</sup>        | 0.923 <sup>4</sup>        | <b>0.031</b> <sup>2</sup> | 0.038 <sup>2</sup>        | <b>0.054</b> <sup>3</sup> | <b>0.008</b> <sup>1</sup> | <b>0.08</b> <sup>1</sup> | <b>0.09</b> <sup>1</sup> | <b>0.60</b> <sup>1</sup> |
| <i>Ohio 2020 / MLP (Ohio 2019 models, no refitting)</i> |                           |                           |                           |                           |                           |                           |                           |                          |                          |                          |
| ERM                                                     | <b>0.830</b> <sup>1</sup> | <b>0.894</b> <sup>1</sup> | <b>0.940</b> <sup>1</sup> | 0.086 <sup>9</sup>        | 0.095 <sup>7</sup>        | 0.143 <sup>7</sup>        | 0.193 <sup>4</sup>        | 80.17 <sup>4</sup>       | 4.53 <sup>7</sup>        | 22.94 <sup>5</sup>       |
| NeuFair                                                 | 0.808 <sup>7</sup>        | 0.880 <sup>8</sup>        | 0.934 <sup>9</sup>        | 0.016 <sup>2</sup>        | <b>0.023</b> <sup>1</sup> | 0.029 <sup>2</sup>        | 0.095 <sup>2</sup>        | 61.18 <sup>2</sup>       | 1.83 <sup>2</sup>        | 31.96 <sup>8</sup>       |
| <b>INTERFAIR</b>                                        | 0.823 <sup>4</sup>        | 0.890 <sup>5</sup>        | 0.939 <sup>6</sup>        | <b>0.010</b> <sup>1</sup> | 0.035 <sup>2</sup>        | <b>0.020</b> <sup>1</sup> | <b>0.008</b> <sup>1</sup> | <b>0.08</b> <sup>1</sup> | <b>0.10</b> <sup>1</sup> | <b>0.47</b> <sup>1</sup> |

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